FREQUENCY RESPONSE AND NODAL MAPPING OF SINGING BOWLS

by

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BRIGHAM YOUNG UNIVERSITY - IDAHO

DEPARTMENT APPROVAL

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ABSTRACT

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Singing bowls are musical instruments which, upon excitation, produce sound. Traditional methods of excitation of singing bowls include striking with a mallet and continuously exciting the bowls using a wooden rod wrapped in suede. The method that is used to excite the bowls to produce sound has a large effect on the frequency and intensity levels of the sounds that are produced by each bowl. Exciting the bowls using traditional methods is explored briefly in this study. A different way of exciting singing bowls is used in this study, and this involved a vibrational generator pressed against the side of the bowl, with a Lock-In Amplifier that had a microphone as an input. This approach to exciting singing bowls and measuring the sound produced opened new avenues of study into the dynamics and oscillatory characteristics of singing bowls. Seven hand-hammered singing bowls were studied, and new characteristics of each

bowl were recorded. The response singing bowls have to a range of frequencies around the harmonic frequencies of the first two harmonic oscillations of each bowl were studied. The frequency response curves that resulted from these studies yielded the quality factor for the targeted resonant frequencies. Exciting the singing bowls using a vibrational generator also allowed for greater insight into how the rim of the bowls deform in order to produce the sounds heard. Singing bowls oscillate in predictable patterns of the shape of the rim of the bowls, forming nodes and anti-nodes. Using a vibrational generator allowed for the formation of standing waves in the rim of the bowl. Then the intensity of the sound at various points around the bowl was measured. The resulting nodes and anti-nodes of the standing waves of the rim of the bowls became visible, and were used to map how the rim of each bowl oscillates and bends in response to the driving force at the resonant frequencies studied.

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Chapter 1

Introduction

Singing bowls are instruments which, upon excitation by various forces, are excited to a range of modes of oscillation, causing the bowls to produce sound. The modes of oscillation of these bowls are responsible for a range of different frequencies of sound. The set of frequencies in the sounds produced by each bowl are dependent upon the size and shape of each bowl, the material of the bowl, the material upon which the bowl is being rested, and upon the method which is used to excite the bowl to oscillate. The frequencies produced by each bowl can be seen distinctly on a frequency-intensity plot. This is known as the frequency spectrum of the bowls. Each peak of these spectral plots corresponds to a different mode of oscillation of the rim of the bowl and how it bends in order to produce each frequency of sound. Each mode of resonance has a quality factor associated with it, which tells how long energy can be stored within that oscillation. In this chapter, these ideas will be elaborated upon. The three goals of this study were to determine how different methods of excitation effect the acoustical properties of seven hand-beaten singing bowls, produce response curves for the first two harmonic oscillations of the singing bowls and determine how well the equation for a one-dimensional driven oscillator fits these curves then

determine a resulting quality factor for the oscillations of the bowls, and lastly to map the nodes and anti-nodes of the first two harmonic oscillations of each of the bowls.

1.1 How Singing Bowls Make Sound

Idiophones are instruments which, when struck, oscillate and produce sound. One common example of an idiophone is a tuning fork, which is a tuned u-shaped piece of metal that produces a longitudinal disturbance in the air at the free end of the tuning fork.[1] Much like a tuning fork, singing bowls are idiophones, and have rims which oscillate back and forth to produce sound.

Singing bowls not only can be struck, but also can be continuously excited in order to produce sound. Traditionally, singing bowls have either been percussed using mallets or continuously excited using a puja. A puja is a wooden rod that is wrapped in suede.[2]

The quality and characteristics of the sound produced by singing bowls are largely dependent on the method that is used to excite the bowls. When struck, the rims of singing bowls deform momentarily and "cycle through" different shapes of the rim, which create the sound that is heard. When continuously excited with a puja by rubbing the puja around the outside of the rim of the bowl, one shape is prominent, and so the sounds produced by the bowl are mostly due to one frequency and shape of oscillation. The angle, pressure, position, and speed that a puja has while it makes contact with the rim of the bowl all determine what shape the rim takes as its most prominent oscillation.[3][4] It is possible to excite some singing bowls to its first and second harmonic while using a puja by changing the angle of the puja and the force applied to the rim.

Each shape of the rim has points of maximum displacement and minimum displacement, otherwise known as anti-nodes and nodes, respectively. The first two sound-producing resonances of a bowl are characterized by four anti-nodes for the first harmonic resonance, and six anti-nodes for the second harmonic resonance.[5] The first and second harmonic resonance will be focused on in this study. These two resonances line up with the first and second peaks on a frequency-spectrum graph. An example of a frequency-spectrum graph is seen below in Figure 1.1.

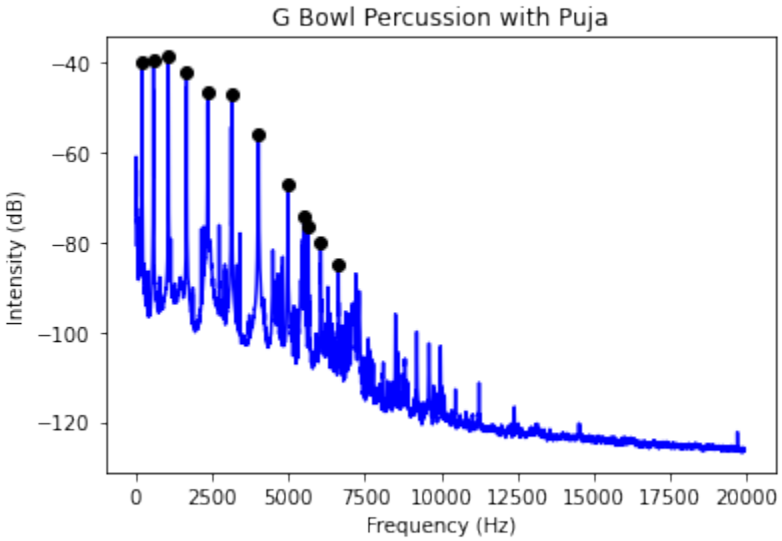

Figure 1.1 Frequency spectrum of singing bowl out to limit of human hearing

Seen above is a spectrum graph of one of the singing bowls in this study, out to the limit of human hearing (20,000 Hz). This graph shows the G-Bowl after it has been struck using a puja. Since only the first two harmonic overtones will be focused on in this study, frequency spectrum graphs have been narrowed to a range of 0 Hz to 7000 Hz in the rest of this study.

1.2 Spectra of Singing Bowls

Frequency-spectrum graphs provide a picture of the spectrum of frequencies each bowl can produce. Each frequency peak on this graph has an associated mode of oscillation, which describes how a singing bowl deforms in order to produce that frequency of sound. Lower frequencies tend to have fewer nodes and anti-nodes. Each harmonic overtone of a singing bowl is a combination of two resonance frequencies within 2-10Hz, and these double overtones can be seen on the frequency spectrum graph as two closely-spaced peaks. This close spacing of resonance generally makes each overtone double-peaked. This close pairing of frequencies at each overtone create a "beating" of the sound produced by the bowls.

1.3 Eliminating Background Noise

When taking acoustical measurements, noises other than what is being measured can impact the quality of the data that is collected. Anechoic chambers can be used to reduce background noise by using noise-absorbing materials in order to isolate the subject and subsequent measurements from outside sources. In a non-anechoic environment, background noises are present, so acoustical measurements are influenced by these outside sources of noise, as long as that frequency of sound is not present in the surrounding environment.

In previous studies, others have used accelerometers and slow-motion imaging to ascertain the dynamics of singing bowls and to get around the obstacle of background noise.[6] In this study, all measurements taken are based solely on acoustical data in a noisy, non-anechoic environment, which means there is a need to isolate the sound of the singing bowls from the surrounding noise. This is done using a Lock-In Amplifier.

If the frequency of sound is known, background noise is able to be reduced sig-

nificantly through the use of a Lock-In Amplifier (LIA). The LIA, along with a microphone as an input, and a reference frequency provided by a waveform generator as a synchronizing waveform, can show how much of a specific frequency of sound is present in the surrounding environment. Basically, it is "locking in" to that frequency, hence the name. LIAs pick up on only the reference frequency provided by the waveform generator at its particular phase, so background noise is minimal as long as the reference frequency is not present in the surrounding environment at the phase of the reference signal.

The LIA is a device which, when coupled with an input of a Waveform Frequency Generator as a synchronizing mechanism, and a microphone as an input, can measure how synchronized that incoming signal is to the waveform generator's frequency and phase. The LIA has display windows for the waveform phase difference, and the V_{rms} which is a measurement of the intensity of the sound that is picked up by the microphone. On the LIA, the intensity level of the incoming sound at the specified frequency and the phase difference are both measurable.

1.4 Response Curves and The Quality Factor

Driven oscillators have a measurement of the quality of the oscillation called the quality factor (q-factor). The q-factor is an indication of how long the oscillation can persist when the driving force is removed.[7] The q-factor for each bowl in this study is examined through driving the bowls at varying frequencies slightly above and below the resonant frequencies. This provides a picture of the response the bowls have to that frequency of excitation. These are called response curves, and can be used to find the q-factor.[8] Below in Figure 1.2 is an example of a response curve of a tuning fork.

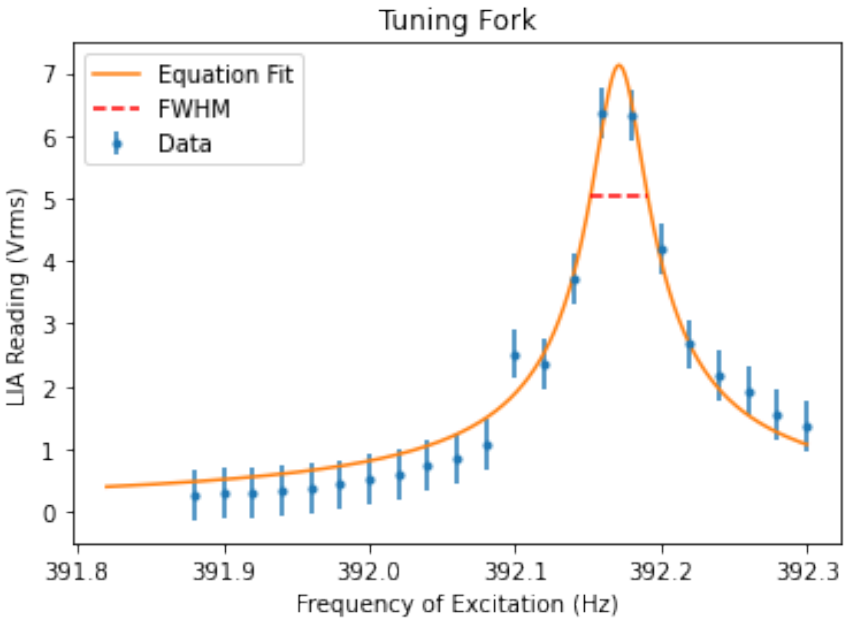

Figure 1.2 Response curve of a tuning fork

This particular fit of a response curve yields a q-factor of 10027.79 ± 0.05. This dimensionless measurement of this oscillator is a measurement of the energy found in the driven oscillations of the tuning fork. This value is indicative of the "strength" of the oscillation that the tuning fork has at the frequency of this particular driven oscillation at a range of frequencies, as the resonance frequency is passed over. Not all graphs are equal, as this study will show. The response curve above was fitted to the equation below in order to get the q factor:

$$|\frac{X}{(\frac{F}{k})}| = \sqrt{\frac{1}{(1-\frac{\omega^2}{\omega_n^2})+(2\zeta\frac{\omega}{\omega_n})^2}}$$

Where X is the response, F/k is the response to a static force, ω is the frequency, ω_n is the resonant frequency, and ζ is the damping coefficient.[9] The above equation describes a driven 1-dimensional oscillator. This study will explore how well the equation for a 1-dimensional oscillator can fit the complex oscillations that occur when singing bowls sing.

Using the fit equation explained above, the q-factor is then determined. Since these graphs are amplitude-based graphs, the full width at half max (FWHM) is taken to be the full width at max amplitude divided by $\sqrt{2}$. The equation to compute the q-factor can be seen below:

$$q = \frac{\omega_n}{\Delta\omega}$$

Where q is the q-factor, ω_n is the resonant frequency, or the peak of the equation fit, and $\Delta\omega$ is the bandwidth at the resonant frequency divided by $\sqrt{2}$. This makes the fit equation for FWHM at around 70% the max amplitude, as opposed to 50% of the max amplitude that other studies have used.[10]

Chapter 2

Procedure

Seven hand-beaten singing bowls were used in this project. Each bowl was tuned to the seven main notes of the chromatic music scale: A, B, C, D, E, F, and G. The bowls, in order of smallest to largest size is: E, D, G, F, C, A, and B. These bowls can be seen below in Figure 2.1.

Figure 2.1 The seven hand-beaten singing bowls used in this study

The diameters of the bowls can be seen below in Table 2.1, in the order of largest to smallest. Measurements were taken at seven evenly-spaced points around the rim. These measurements were averaged, and standard deviations are given.

Table 2.1 Average Bowl Diameters

Bowl:	B-Bowl	A-Bowl	C-Bowl	F-Bowl	G-Bowl	D-Bowl	E-Bowl
Average Diameter (cm)	27.53	25.84	24.20	22.03	18.57	16.87	14.26
Standard Deviation (cm)	0.08	0.10	0.18	0.08	0.08	0.15	0.05

Since just the diameter of the bowls does not give an accurate size of the acoustical chamber of the bowls, the depth of the bowls is also needed. Measurements of the depth of the bowl were taken at seven points around the rim. Averaged bowl depths along with the standard deviation can be seen below in Table 2.2.

Table 2.2 Average Bowl Depths

Bowl:	B-Bowl	A-Bowl	C-Bowl	F-Bowl	G-Bowl	D-Bowl	E-Bowl
Average Bowl Depth (mm)	122	113	104	103	90	86	75
Standard Deviation (mm)	0.5	1.3	1.1	1.1	0.4	1.7	1.7

The weights of the A, C, D, E, F, and G bowls were measured using a hanging package scale which had an uncertainty of 25g. The weight of the B bowl was measured using a OHAUS hectogram hanging scale due to the B bowl exceeding the weight capacity of the hanging package scale. This scale had an uncertainty of 250g. The bowls were placed in a plastic bag, and hung from the respective scales. Bowl weights can be seen below in Table 2.3.

Table 2.3 Bowl Weights

Bowl:	B-Bowl	A-Bowl	C-Bowl	F-Bowl	G-Bowl	D-Bowl	E-Bowl
Bowl Weight (g)	2750	1800	1875	1500	975	800	700

Using an X-Ray Fluorescence device, the composition of the smallest bowl (E-Bowl) was measured. The bowls were found to have a composition of Copper, Iron, Antimony, Zinc, and Nickel. These metals and their abundance in the bowls are what give the bowls their acoustical qualities.

In this study, base measurements of the frequency spectra of singing bowls were measured using the traditional methods of striking using a puja and mallets, and continuous excitation using a puja. In addition, the bowls were continuously excited using a vibrational generator. Using the vibrational generator, frequency-response

curves of the first two harmonic oscillations for each of the seven bowls were experimentally acquired. After this, the nodes and anti-nodes of the first two harmonics of each of the bowls were acoustically mapped. The process of experimental setup and procedure are explained in the following sections.

2.1 Frequency Spectrum of Singing Bowls using Traditional Methods

The singing bowls were set on round padded bases, and tape was used to secure the bowls to the base, and to secure the padded base to the table. The tape was necessary, since without it the bowls would wiggle around when struck with the mallets. A Zoom H6 field recorder was set up 6cm ± 0.5cm away from the rim of the bowl, with the microphones of the recorder being pointed toward the rim of the bowl. An example of this setup can be seen below in Figure 2.2.

Figure 2.2 Striking experimental setup

Striking measurements were taken for each of the 7 bowls using a small puja with a diameter of 2.6cm ± 0.1cm, a small mallet with a diameter of 6.0cm ± 0.1cm, and a large mallet with a diameter of 7.1cm ± 0.1cm. The bowls were struck on the side of the bowl 10-15 times for the measurements taken with the puja and both mallets. After each strike, the bowls were allowed to ring for 3-5 seconds before they were silenced by touching a finger to the rim of the bowl.

Audio measurements were also taken while the bowls were being continuously excited using the large and small puja with a diameters of 4.2cm ± 0.1cm and 2.6cm ± 0.1cm. To set up the continuous excitation experiments, each bowl was held in one hand with the bowl resting on fingertips, and the puja was used to excite the bowl by rubbing the outside of the rim of the bowl in a circular motion with the puja in contact with the outside of the rim, exciting the bowl to a steady state of sound. An example of this setup is given below in Figure 2.3.

Figure 2.3 Continuous excitation with a puja experimental setup

For the continuous excitation experiments, the Zoom H6 recorder microphones were set up around 20 cm ± 3cm. This large uncertainty was due to unavoidable movement as the bowl was played. The bowls were played for 10-15 seconds once a steady tone was produced. Both the small and large puja were used to excite the bowls in this manner.

Some bowls were able to be excited to their second overtone with the large puja. This was accomplished by making the angle the puja made with the rim larger, by applying more pressure to the rim, and by increasing the speed at which the puja moves around the bowl. The second overtone measurements while the bowl was being continuously excited with the large puja were taken for the F, C, A, and B bowls. The G, D, and E bowls were not able to be excited to the second overtones with the large or small puja.

2.2 Response Curves Using Lock-In Amplifier

Each of the singing bowls were excited under a driven oscillation. This process will be described in this section. Three felt pads were stuck on the bottom of the bowl to support the bowl and to lift the bowl off of the table. Two pads were used on the end of the vibrational generator rod where the rod of the vibrational generator made contact with the bowls. A Keysight 33600A Series Waveform Generator and a Stanford Research Systems SR530 Lock-In Amplifier were used in this study. A model 426E01 ICP 040802 PCB Piezotronics Condenser Microphone was connected to a model 480C02 ICP Sensor Signal Conditioner, which was then connected to the LIA input 1. The waveform generator was set as the output for the vibrational generator. Specifics about this setup are described in the next few paragraphs. A schematic of this setup can be seen below in Figure 2.4.[10]

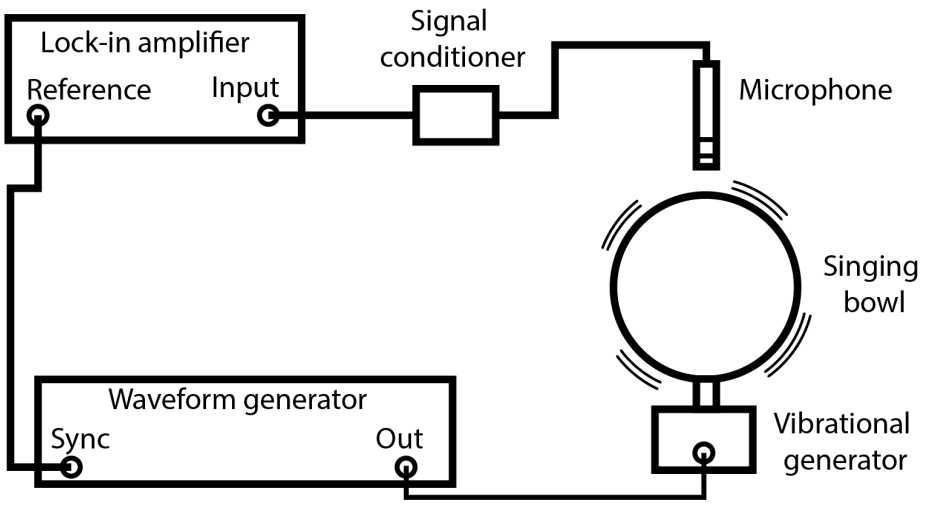

Figure 2.4 Schematic of experimental setup used for LIA-based experiments

The waveform generator was set to an intensity level of 1.5 Volts Peak-to-Peak (V_{pp}), and was set at the same frequency the LIA was picking up on. The LIA was set to a sensitivity level ranging from 10 to 50 mV, depending on the intensity of the sound being produced by the bowls. The Dynamic Resolution setting on the LIA was set to Norm, and the Time Constant setting was set to 300 ms. Uncertainty in the measurement of the intensity readings by the LIA was found using an oscilloscope. This uncertainty value was measured after all of the experiments were done using the F-bowl while it was being excited to its first harmonic resonance at a LIA sensitivity of 10 mV. The uncertainty in the measurement of the LIA was determined to be 0.4 mV.

The tip of the vibrational generator was set to touch the side of the bowls, at the point where the bowl curves up to form the rim. The place where the vibrational generator tip was set on the bowls was at a varying distance beneath the rim on each bowl, as each bowl had a differently shaped rim. The positions of the contact points of the vibrational generator that was used for each bowl can be seen in Appendix A, Figures A.4 through A.8. Some bowls needed to be set on a block of wood in order

for the vibrational generator tip to reach the correct position on the side of the bowl. The microphone was positioned approximately parallel to the rim of the bowls, at a height matching that of the rim. Below in Figures 2.5, 2.6, and 2.7 is an example of this experimental setup.

Figure 2.5 Felt pad setup on bottom of bowls

Figure 2.6 G-Bowl vibrational generator setup

Figure 2.7 Microphone vibrational generator setup

A point of maximum intensity of sound was found using the waveform generator through incrementally increasing or decreasing the diving frequency of the vibrational generator until a maximum V_{rms} reading was reached on the display of the LIA. The phase difference reading on the LIA was zeroed by pressing the REL button near the phase difference reading window on the LIA. Then, the driving frequency of the waveform generator was decreased until the phase difference reading on the LIA was at around -90 degrees. The waveform generator was then used to go back up to, through, and past the resonance frequency, up to +90 degree reading on the LIA phase difference window. The driving frequency was progressed in 0.02 - 0.10 Hz increments. The two partials of the first two resonant harmonics of each bowl were studied in this manner. Values of the intensity, or V_{rms}, was recorded after each progression of driving frequency. The phase difference was also recorded for each incremented step.

2.3 Nodal Mapping Using Lock-In Amplifier

To map the nodes and anti-nodes of each bowl, the microphone was incrementally moved around the bowl, and every 10 degrees the intensity of the sound was measured. The microphone was set up 2cm ± 0.5cm away from the rim of the bowl. The rim and the top of the microphone were set at the same level, and the microphone was positioned and angled so that the face of the microphone was about perpendicular to the rim of the bowl. Below in Figure 2.8 is an example of the setup for this experiment.

Figure 2.8 Nodal mapping setup

2.4 Data Analysis Techniques

The audio files obtained using the Zoom H6 Field Recorder were imported into the computer program Audacity, and a Fast Fourier Transform was performed on the

audio file. This was used to obtain the Frequency-Spectrum plots. This data was then exported as a .txt file, and a Python program was written to find the peak resonances of each overtone. An example of the Python code used can be seen in Appendix A.

The response curves were also analyzed in Python, and the data was fitted to a curve. Initial guesses for the function parameters were determined through fitting the one dimensional oscillator equation to each response curve using the program LoggerPro. The Full Width at Half Maximum (FWHM) was then found. This value allowed the q-factor to be calculated. A reduced chi squared test of the fit of the data to the function described at the end of the previous chapter for a 1-dimensional oscillator was also performed. An example of this code can be seen in Appendix A.

The LIA displays values of V_{rms} in mV. This is able to be converted into dB, the Sound Pressure Level (SPL) reading for the intensity of sound. The sensitivity of the microphone used is needed for this, and it was found to be 50 mV/Pa. Below is the process by which this conversion from Vrms to SPL was done:

$$P_{rms} = V_{rms} * \frac{1\text{Pa}}{50\text{mV}}$$

$$SPL = 20\text{dB} * \text{Log}_{10}(\frac{P_{rms}}{20\mu\text{Pa}})$$

Where P_{rms} is the pressure of the sound wave picked up by the microphone. The microphone used had a calibration factor of 50 mV/Pa. Pa is Pascal, meaning pressure. SPL is the Sound Pressure Level in dB. 20 μPa is the reference pressure for SPL.

The Reduced Chi Squared tests of fit were performed using the following equation:

$$\chi^2 = \frac{1}{N-2}\sum\frac{(\omega - f(x,\omega_{fit}))^2}{(\sigma)^2}$$

Where N-2 is the degrees of freedom, ω is the V_{rms} value of the individual data points, f is the equation fit described in the first chapter, x is the x value of each data point, ω_{fit} is the value of the equation line at each data point, and σ is the uncertainty in the LIA measurement.[11]

Chapter 3

Results

The research done in this study was quite extensive. The A-Bowl, C-Bowl, and D-Bowl were chosen to be previewed in this section. The full array of results for the all of the experiments can be seen in Appendix B.

3.1 Traditional Methods

The results for the striking and continuous excitation experiments for the C-Bowl can be seen below (Figures 3.1 through 3.6). The range of these graphs were narrowed down to between 0 Hz and 7000 Hz. Additional frequency spectrum graphs can be seen in Appendix B. In this study, three means of exciting the bowls were studied. Classical methods of excitation included striking experiments while using both a Puja and a mallet, and continuous excitation experiments were performed using the small and large pujas. These experiments provide initial guesses for the resonant frequencies of the bowls for the response curve experiments.

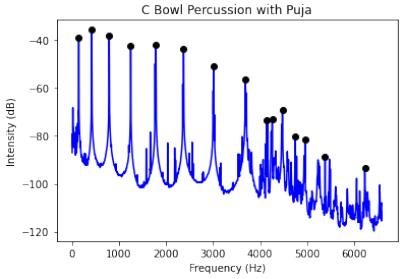

Figure 3.1 C-Bowl frequency spectrum after striking with small puja

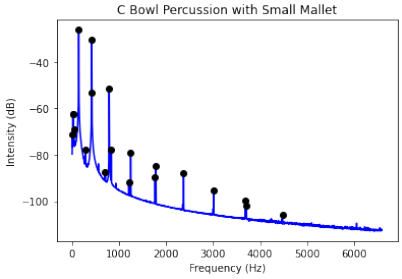

Figure 3.2 C-Bowl frequency spectrum after striking with small mallet

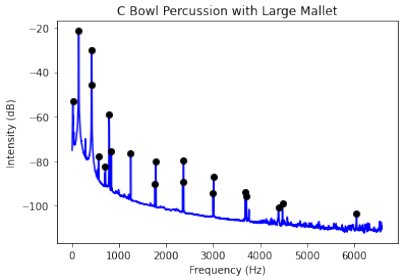

Figure 3.3 C-Bowl frequency spectrum after striking with large mallet

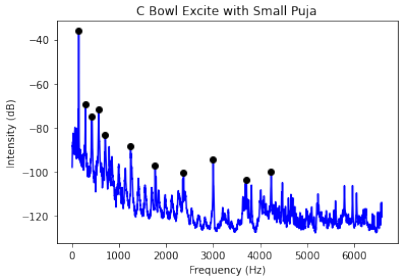

Figure 3.4 C-Bowl frequency spectrum while continuously exciting with small puja

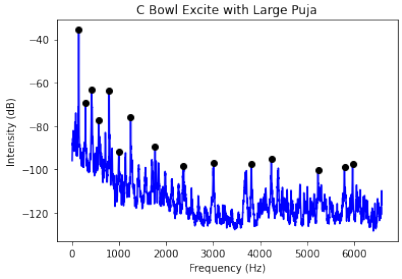

Figure 3.5 C-Bowl frequency spectrum while continously exciting with large puja

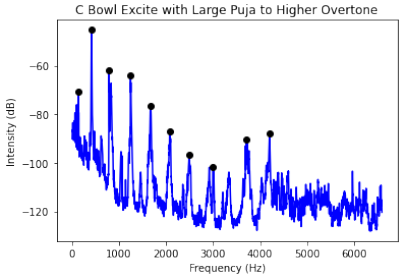

Figure 3.6 C-Bowl frequency spectrum while continuously exciting to second overtone with large puja

3.2 Response Curves and Q-Factors

Below is the intensity reading of the sound as the response curve was performed, and the phase difference reading of the LIA (see Figure 3.7). The A Bowl had paired resonance frequencies close enough together that both resonances were able to be captured in the same graph. Beside this graph is a graph of the equation fit for the A Bowl. The equation mentioned in Chapter 1 did not fit correctly (see Figure 3.8), so the resonance pairs were split up and analyzed separately (See Figures 3.9 and 3.10). The two partials for the second overtone for the A bowl can also be seen below in Figures 3.11 and 3.12. The uncertainty value determined by an oscilloscope was found to be 0.4 mV, and this is what was used to make the error bars in each of the response curve graphs. Reduced chi squared fits for each graph are given at the table at the end of this chapter, and below each graph. A full array of results for the response curve experiments can be seen in Appendix B (Figures B.55 through B.82

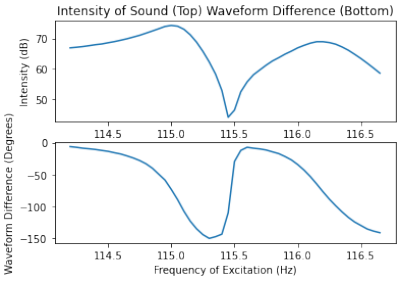

Figure 3.7 A-Bowl intensity and phase difference

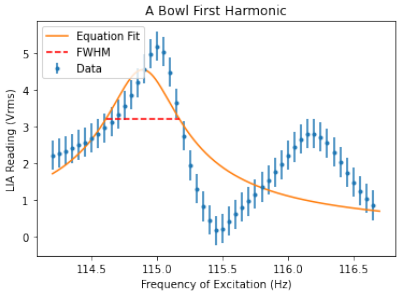

Figure 3.8 A-Bowl bad equation fit response curve

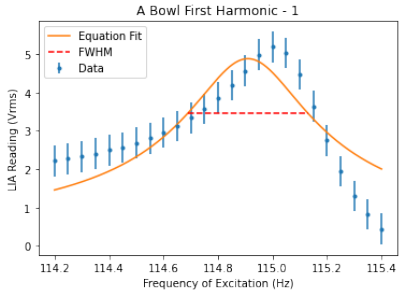

Figure 3.9 A-Bowl response curve first harmonic first partial. RChiSq: 3.15

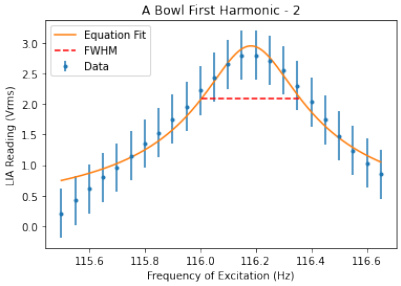

Figure 3.10 A-Bowl response curve first harmonic second partial. RChiSq: 0.23

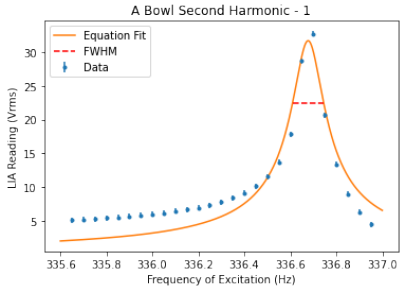

Figure 3.11 A-Bowl response curve second harmonic first partial. RChiSq: 40.6

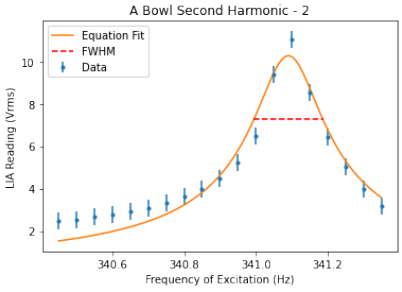

Figure 3.12 A-Bowl response curve second harmonic second partial. RChiSq: 2.56

The C Bowl had a very unique response curve. The response curve protocol was ran 2 times, with a different yet similar set up each time, for this bowl, with approximately the same result which is seen below in Figures 3.13 and 3.14. The response curve had to be divided into 2 separate curves, as was with the case of the A-Bowl response curve. The division can be seen in Appendix B, Figures B.62 and B.63.

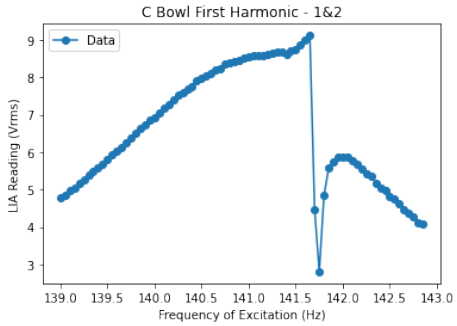

Figure 3.13 C-Bowl first harmonic first and second partial

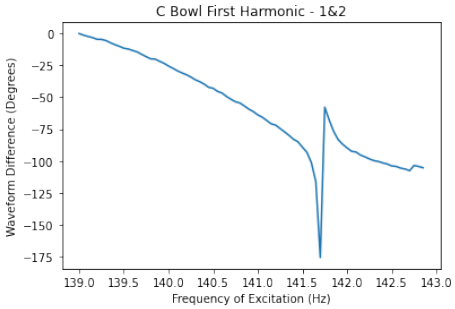

Figure 3.14 C-Bowl response curve waveform difference

Below (Table 3.1) is a table of the quality factors associated with the two partials of the first two harmonic frequencies for each bowl. Each row is labeled with the harmonic overtone number separated by a dot from the number of partial of the harmonic. I.e. 1.2 means the first harmonic overtone and the second partial of the harmonic. A full array of the response curves that led to these q-factors can be seen in Appendix B.

Table 3.1 Q-Factors

Harmonic.Partial	B-Bowl	A-Bowl	C-Bowl	F-Bowl	G-Bowl	D-Bowl Thick Side	D-Bowl Thin Side	E-Bowl
1.1	68.11	259.79	50.60	86.02	436.2	1191.92	622.47	886.26
1.2	75.02	326.08	119.54	-	126.03	705.69	508.74	-
2.1	3312.27	2473.44	2112.06	3560.92	3718.63	8221.07	8502.92	12430.00
2.2	-	1769.60	-	7292.66	11103.37	5994.05	4329.27	8421.42

Seen below in Table 3.2 are the reduced chi squared values for the fits of the

equation mentioned in Chapter 1 for the response curves performed. Each row is labeled with the harmonic overtone number separated by a dot from the number of partial of the harmonic. I.e. 1.2 means the first harmonic overtone and the second partial of the harmonic. The reduced chi squared values for each graph are also printed below each response curve figure in Appendix B.

Table 3.2 Reduced Chi Squared Values for Response Curve Equation Fits

Harmonic.Partial	B-Bowl	A-Bowl	C-Bowl	F-Bowl	G-Bowl	D-Bowl Thick Side	D-Bowl Thin Side	E-Bowl
1.1	0.06	3.15	1.76	6.95	1.57	5.28	1.61	1.84
1.2	0.83	0.23	1.52	-	1.45	4.87	1.33	-
2.1	3.89	40.6	0.17	1.25	3.79	0.56	0.18	0.40
2.2	-	2.56	-	13.2	0.56	0.5	0.27	0.11

3.3 Nodal Mapping

The first and second harmonic overtones were found and the nodal mapping experiments were run for each bowl. The results of the D-Bowl can be seen below for the first two harmonic overtones (Figures 3.15 through 3.17). Due to the large difference between the two partial frequencies of the first harmonic overtone, both partials were mapped. For the rest of the bowls, only one of the partial frequencies were mapped. A full listing of the results can be seen in Appendix B.

For further clarification, below in Figure 3.18 are predictions obtained from a previous study which detailed the nodes and anti-nodes present in the different harmonic overtones of singing bowls.[5] This figure contains the first 16 overtones of the oscillations of the rims of the singing bowls. It is helpful to compare the first two overtones in this study to that of the study pertaining to this figure.

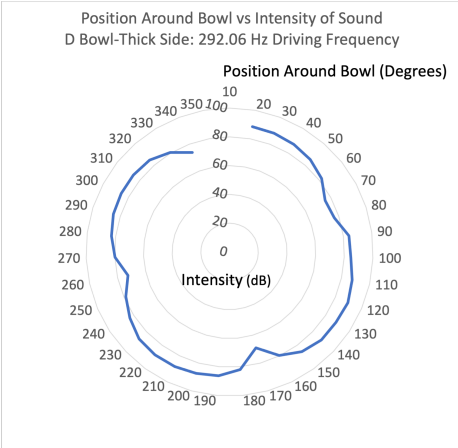

Figure 3.15 D-Bowl thick side first harmonic first partial nodal map

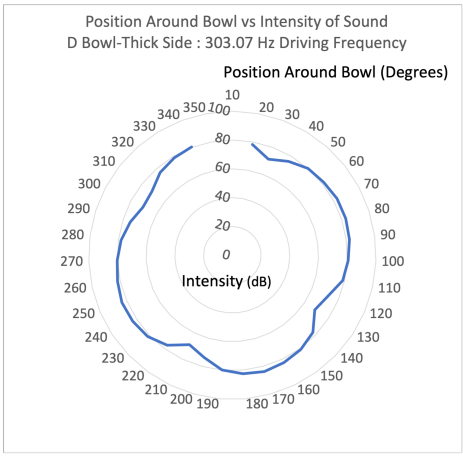

Figure 3.16 D-Bowl shick Side first harmonic second partial nodal map

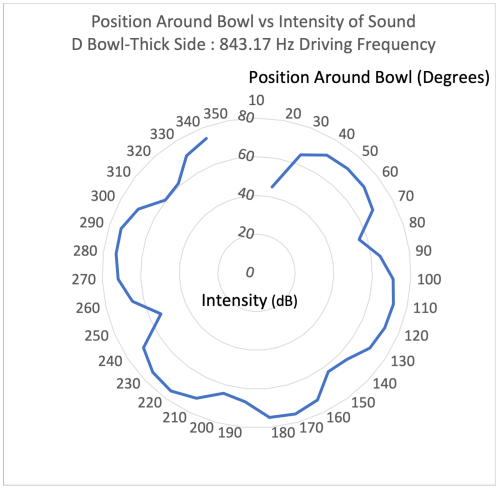

Figure 3.17 D-Bowl thick side second harmonic nodal map

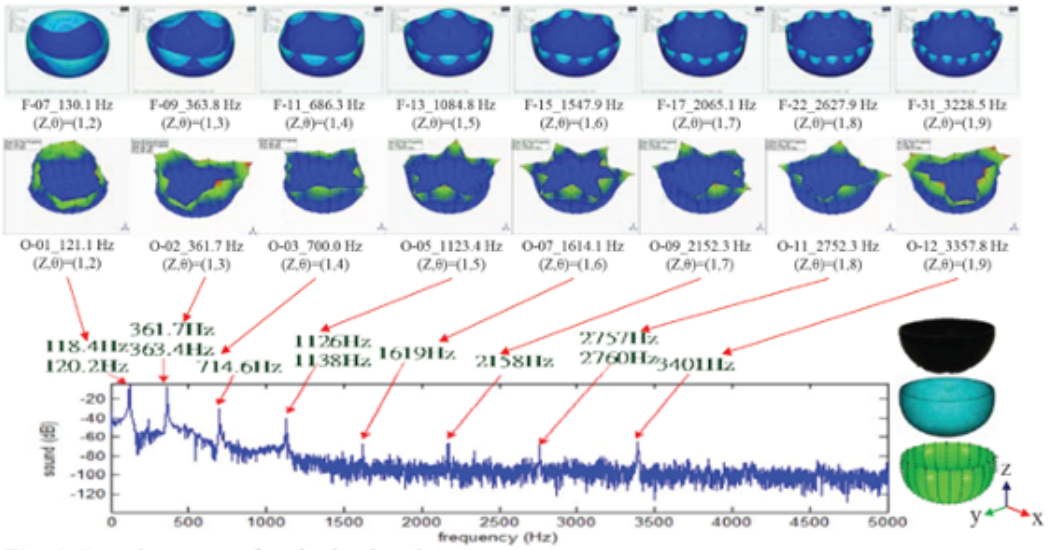

Fig. 4. Sound spectrum for singing bowl A.

Figure 3.18 Wang *et al.* nodal plots

Chapter 4

Discussion

The methods of excitation of the bowls using traditional methods (striking and continuous excitation) reveal the range of frequencies each bowl can produce. The frequency spectrum graphs seen in Appendix B (Figures B.1 through B.40) identify what frequencies the bowl is sensitive to, and what sounds each individual bowl produces. These frequencies were used to determine what frequencies each bowl should be excited at with the vibrational generator for the other parts of the experiments.

With the striking experiments, as can be seen from the frequency spectrum graphs mentioned before in Appendix B, the instrument used to strike the bowl had a large effect on what the frequency spectrum looked like for each bowl. Using the puja to strike the bowls produced a broader array of frequencies at a higher intensity for more overtones than striking with a mallet did. Striking a bowl with a small mallet with less mass produced a slightly different spectrum than striking with a large mallet that had more mass did. Mainly this difference can be seen in the closely-spaced overtones on each resonant harmonic. Some of the partials were at a lower intensity within these pairs of overtones, changing the overall tone the bowl produced when it was struck.

The D-Bowl had structural inconsistencies that made it have partial overtones much further away from each other than the other bowls. One side of the bowl had a rim thickness of 4.91mm, while the opposite side of the bowl had a rim thickness of only 2.42mm. This made the bowl have partial frequencies for the first overtone of around 292Hz and 301Hz. This difference of almost 10Hz was much larger than the other bowls. Most of the other bowls had a difference between partials within the first and second overtones of only 2-3Hz.

The continuous excitation experiments had a range of outcomes, based on the size of the bowls. Continuous excitation with a puja variously excites and dampens the oscillations of the bowl. The B-Bowl, as can be seen in Figure B.10, was not able to be excited very well using the small puja, causing the sounds produced by the bowl to be not intense and almost inaudible to the human ear. The reason for this has been explored by others in the past, and has to do with the diameter of the puja and incompatible stick-slip interactions while the puja is traveling around the rim of the bowl.[4] The B-Bowl was able to be excited with the large puja to an audible intensity level. The D-Bowl was not able to be continuously excited using the small nor large puja due to the structural deformities mentioned before. All of the other bowls were able to be continuously excited using the small and large puja. Only the A, B, C, and F Bowls were able to be excited to the second overtone during the continuous excitation experiments. This is presumably due to their larger sized acoustic chambers, allowing them to accommodate the higher mode of resonance while being continuously excited with a puja.

During experiments using the vibrational generator, audio was recorded, and frequency spectrum graphs for the first two overtones of the bowls can be seen in Appendix B (Figures B.41 through B.54). As can be seen from these frequency spectrum graphs, the harmonic overtone of the bowls (the first and second overtones) were ex-

cited much more than other overtones, as was expected. What was surprising; however, was that while the bowls were being excited to their first harmonic overtone, the second harmonic overtone was often present. During excitement to the second overtone, it was common for other, higher overtones to also be present in the frequency spectrum graphs. The D-bowl continued its trend of being an outlier, and had a frequency spectrum graph that displayed sporatic peaks of varying frequencies, which was not seen in other frequency spectrum graphs taken during these experiments for other bowls.

Seen in Figures 3.9 through 3.12 are the response curves of the A-Bowl. Response curves for the rest of the bowls can be seen in Appendix B (Figures B.55 through B.82). Some bowls were not able to be excited to both its first and second partial overtones for the first and second harmonic on the singular trial that was ran with the particular response curve setup, but some were. These response curves yield a q-factor, which gives a picture of how the sound dissipates with respect to time. A lower q-factor means that the sound produced will often be of greater intensity, and therefore more audible. The q-factors obtained from the response curves that were listed in previous chapter are incomplete, as some response curves were not able to be performed, presumably due to the setup of the experiment. More data is needed to ascertain a good q-factor value for these partial overtones.

A rather high value for the q-factor is around 1000, so the q-factors obtained in this study were peculiar. The equation for the curve fit did not fit well to some response curves, causing the resulting q-factor to not be accurate. The reduced chi squared values seen in Table 3.2 for each response curve indicate that, for the most part, the equation for a 1-dimensional oscillator is a good fit to model the complex oscillations of the singing bowls, since the reduced chi squared value was under 1. However, for some graphs, the equation did not fit well. This was generally seen when the response

curve for the oscillation was very sharp. Modification of the 1-dimensional oscillator equation, or better means of identifying the initial guesses for the fit of the equation to the data, is potentially needed to gain a better fit of the equation to these unique response curves.

The response curves were very sensitive to small changes in the experimental setup. The spot where the tip of the vibrational generator touched the bowl had a large effect on how well the bowl was able to oscillate and therefore had a large effect on the intensity of the sound produced. The point on each bowl where the maximum sound was produced was found, and the setup used for each bowl can be seen in Appendix A (Figures A.4 through A.8).

Nodal mapping highlights a few different characteristics of the bowls. The thickness of the rim, the depth of the bowl, and the varying diameters of the bowls all contribute to nodal points that were found often to be not symmetric. These nodes and anti-nodes were mapped for the first and second overtones of each bowl. In the previous chapter, Figures 3.15 through 3.17 depict the nodes and anti nodes as the thick side of the D-Bowl was being excited to its first and second overtones. Since the D-Bowl had such a gap between the two partial overtones within each harmonic overtone, the nodes and anti-nodes of both partials were taken. For the rest of the bowls, only one of the partials of each overtone were visualized. The rest of the nodal maps for each bowl can be seen in Appendix B (Figures B.83 through B.100). One thing to note about the nodal maps of the D-Bowl was that, for the first overtone, the first and second partials had slightly different patterns of the positions of the nodes and anti-nodes. This is presumably an indication of which side of the bowl is responsible for the frequency of sound produced. For the thick side visualized in the previous chapter, the second partial has the vibrational generator at the position of an anti-node. It is hypothesized that this means that the thick side is responsible for

the second partial that had a frequency around 302 Hz, while the thin side was responsible for the first partial that had a frequency around 292 Hz. More experiments are required to solidify this claim.

Chapter 5

Conclusion

The spectra of frequencies produced by each bowl was determined, and this aided in the other parts of this study. All bowls except for the B-Bowl were able to be excited to a steady state of sound using both the small and large pujas. The F, C, A, and B bowls were able to be excited to second overtone while the bowl was being continuously excited with the large puja. The G, D, and E bowls were not able to be excited to the second overtones with the large or small puja.

All bowls were able to be excited to their first and second harmonic overtones while using the vibrational generator, though some partial overtones were not able to be excited, presumably due to the experimental setup. The responses of the bowls when excited to frequencies around the harmonic frequencies were studied, and response curves were made for all seven bowls. Each set of data was fit to the equation for a one dimensional driven oscillator, and using this curve a q-factor was calculated for each partial frequency. These q-factors need further experiments to provide accuracy of the q-factors obtained. Initial trends suggest that the second overtone has a much higher q-factor than the first overtone for all bowls.

The uncertainty in LIA measurement were done on only one of the harmonic res-

onances of one of the bowls (the F-Bowl). This uncertainty was taken to be constant throughout the experiments. If the experiments were repeated, the oscilloscope would be used during each measurement, since some of the frequency response curves had to be performed at a higher sensitivity reading than the 10 mV sensitivity used while determining the uncertainty value of the LIA. This more accurate measurement of the uncertainties of the LIA measurements would possibly make the Reduced Chi Squared values for the fit of the equation for a one dimensional oscillator more accurate.

The nodes and anti-nodes of the first two harmonics of each bowl were acoustically mapped, and the number of nodes and anti-nodes match what has been found in previous studies.[2][5] These nodes and anti-nodes provide a picture of how the bowls oscillate, and are dependent on the overtone which the bowls are being excited to. The nodal mapping of all seven bowls were able to be performed.

One of the biggest successes of this project was the fact that meaningful data was able to be collected in a noisy, non-anechoic environment. Acoustical data is sensitive to background noise, so the data collected in this study is significant in highlighting the variability and importance of the LIA setup. Data was able to be collected from a microphone even with background noise being present, and this setup can be used in other areas to obtain accurate measurements and overcome this background noise hurdle.

The reduced chi squared measurements for the driven 1-dimensional oscillator equation fits for this study suggest that this equation can sometimes accurately model even the complex oscillations found in singing bowls. This being said, the equation fits for some of the more unique response curves yielded a reduced chi squared value much larger than 1. This indicates either that the 1-dimensional driven oscillator equation used needs further refinement to fit the response curves for singing bowls, that the initial guesses while computing the equation fit needs further tuning, or

that the error used in the reduced chi squared computation was too small. In any case, the findings of this study are inconclusive on this matter, and further studies are needed in order to provide a definite answer on whether or not a 1-dimensional driven oscillator equation is sufficient to accurately describe the complex oscillations of singing bowls.

One inconsistency in this study was related to how hard the tip of the vibrational generator was pressed against the side of the bowl. In future studies, a force sensor pad could be used to provide more consistency in the experimental setup to make the force more consistent from bowl to bowl. This could lead to a better understanding of how the force used while driving the bowls to oscillate using a vibrational generator effects the oscillatory capabilities and intensity of sound produced by the singing bowls.

In summary, the goals of this study were achieved. Differences in acoustical properties of each bowl under varying excitation methods was shown. Also, response curves were accomplished for the first two overtones of each bowl, and q-factors for each of the two overtones was calculated. The reduced chi squared fits for each bowl show that the equation for a one-dimensional driven oscillator can describe even the complex oscillations that singing bowls undergo in most cases. Finally, the nodes and anti-nodes for the first two harmonics of all of the seven singing bowls were mapped, and these plots match the numbers of nodes and anti-nodes that other studies have achieved.

Chapter 6

Future Work

As has been discussed, further response curve experiments are needed to provide a more accurate depiction of the q-factor of the bowls. Some response curves fit the equation used well, while other response curves were not able to be fit to the equation accurately. An equation that fits both partials of each overtone at once would be a good start to finding a solution to this problem.

Through experimenting with the bowls, it was discovered that the contact points on the bottom and sides of the bowl can effect the sounds produced by the bowl. As the contact points move from the bottom of the bowl to points further along the side of the bowl, the sound of the bowl was observed to change to a higher overtone during continuous excitation with a puja. A "combination" of overtones was also observed, where both overtones had a similar intensity of sound. Experiments into how the bowls react to differing contact points could lead to discovering more about how the entire bowl oscillates, as opposed to just the rim as this study did.

3-D scanning these bowls was one goal of this project. However, it was not able to be performed due to booking issues. A 3-D scan would lead to a good framework to make a computer model of these bowls, and how the hand annealing process made

them into the instruments that they are. This would help in determining contact points for the experiment previously mentioned in the paragraph above.

The q-factor is largely effected by the surrounding air. If performed in a vacuum, it is expected that the q-factor would differ. This could be an interesting avenue to explore.

Another interesting prospect is to investigate the phase difference. The phase difference yields a factor known as the damping coefficient. Though not focused on in this study, the damping coefficient of the bowl for targeted resonant frequencies could shed further light into how the bowls oscillate.

The health benefits of singing bowls have been studied, and health benefits range from increased vascular performance to increased mental well being.[12] A study of how driven oscillations of singing bowls and the more focused sound produced effects the human body would be beneficial to explore.

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Appendix A

Experimental Setup and Code

In this Appendix A, the setup of the vibrational generator tip on the side of the bowls are pictured. A portion of the code used for analysis has also been included. This code describes how the frequency spectrum graphs were made and how the response curve fit, q-factors, and reduced chi squared tests were performed.

A.1 Vibrational Generator Contact Points

Figure A.1 A-Bowl vibrational generator setup

Figure A.2 B-Bowl vibrational generator setup

Figure A.3 C-Bowl vibrational generator setup

Figure A.4 D-Bowl thin side vibrational generator setup

Figure A.5 D-Bowl thick side vibrational generator setup

Figure A.6 C-Bowl vibrational generator setup

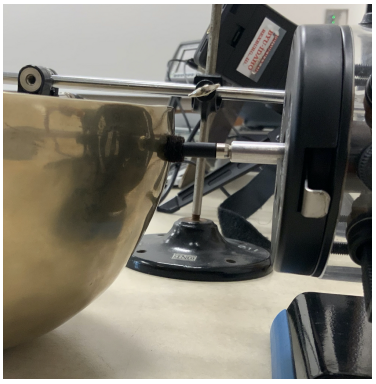

Figure A.7 F-Bowl vibrational generator setup

Figure A.8 G-Bowl vibrational generator setup

Figure A.9 Microphone vibrational generator and nodal mapping setup

A.2 Code Used In This Project

Part 1: Program to plot frequency spectrum and find peaks. Part 2: Program to plot response curves and determine q factor, as well as perform a chi squared test on the fit of the function.

```
from pylab import xlabel, ylabel, title, show, plot, legend, xlabel,
ylabel, title
from scipy import zeros, exp, pi
from scipy.signal import find_peaks
from numpy import loadtxt


#e bowl percussion using puja with ring plus tape


#import FFT frequency analysis size 65536
data5=loadtxt("ePerSP.txt", float)


#frequency
FePer = data5[:,0]
#intensity (dB)
dePer = data5[:,1]


#prominence to weed out insignificant peaks
peaksePer,_=find_peaks(dePer[0:4500], prominence = 20)


plot(FePer[0:4500],dePer[0:4500], "-b")
plot(FePer[peaksePer],dePer[peaksePer],'ko')


title("E Bowl Percussion with Small Puja")
```

```
ylabel("Intensity (dB)")

xlabel("Frequency (Hz)")

show()


print("Frequencies of Partials")

print(FePer[peaksePer])

print("Intensity of Partials")

print(dePer[peaksePer])


from scipy.optimize import curve_fit, fsolve

import pandas

import numpy as np

import matplotlib.pyplot as plt


#program to plot response curve, determine the q-factor,

#and perform a Reduced Chi Squared test on fit


#file name

file = "Avg1.1.xlsx"


#read file

file = pandas.read_excel(file)


#assign collumn names to variables

freqExc = file['Frequency Excite (Hz)'] #Frequencies of excitation

degDiff = file['Waveform Degrees Difference'] #waveform difference

#between waveform generator and sound picked up by the LIA

LIA = file['output reading (mV)'] #LIA reading
```

```
SPL = file['SPL (dB)'] #intensity of sound


#optional plots to get subplots of the response curve
#and the waveform degrees difference plots
fig, ax = plt.subplots(2,1)
ax[0].plot(freqExc, SPL)
ax[1].plot(freqExc, degDiff)


plt.show()


#plt.plot(freqExc, LIA, 'o')


#function to fit data to
def fit(x,f,d,A):
    return A/(np.sqrt((1-x**2 / f**2)**2 + (2*d*x/f)**2))


#range of x values to plot the fit function over
xlins = np.linspace(114.2,115.4,1000)


#Fit for the best fit line for the functions. Parameters
#of max frequency, d, and A are put in as guesses.
#guesses were obtained from values found using Logger Pro.
opt_fit,covfit = curve_fit(fit,freqExc,LIA,p0=[115.5,-0.003523,0.0002583],maxfev=100000)


#Peak value
w0 = max(fit(xlins,*opt_fit))


#peak / sqrt 2
```

```
hm = w0/np.sqrt(2)



#print(opt_fit)

#print(*opt_fit)

#print(covfit)


#print(w0)

#print(hm)


#shifted fit to find roots

def shiftedf(x):

    return fit(x, opt_fit[0], opt_fit[1], opt_fit[2])-hm


#root 1 and 2 are the points where the fitted function meets hm

root1 = fsolve(shiftedf, opt_fit[0]-.1)

root2 = fsolve(shiftedf, opt_fit[0]+.1)


fwhm = abs(root2-root1)


#print(root1,root2,fwhm)


q = opt_fit[0]/fwhm

print("Q-Factor:",q)



point1 = [root1,hm]

point2 = [root2,hm]
```

```
xvals = [point1[0],point2[0]]

yvals = [point1[1],point2[1]]


#Calculate the Reduced Chi Squared for the fit:

#equation at the data points

p = fit(freqExc,*opt_fit)

u = 0.4 #uncertainty in measurement

chisqr = sum(((LIA-p)**2)/u**2) #chi squared

dof = len(freqExc)-2

rchisqr = chisqr/dof

print("rcs:", rchisqr)


plt.errorbar(freqExc, LIA, fmt = '.', yerr = 0.4, label="Data")

plt.plot(xlins,fit(xlins,*opt_fit), label="Equation Fit")

plt.plot(xvals,yvals,"r--", label="FWHM")

plt.legend(loc='upper left')


# Add title and axis names

plt.title('A Bowl First Harmonic - 1')

plt.xlabel('Frequency of Excitation (Hz)')

plt.ylabel('LIA Reading (Vrms)')


plt.show()
```

Appendix B

All Singing Bowl Graphs

In this Appendix B, there is a complete set of all graphs made in this study. Blank spaces are there on purpose, and represent a harmonic overtone partial or graph that was not able to be collected during the course of this study.

B.1 Frequency Spectrum Graphs for Traditional Methods

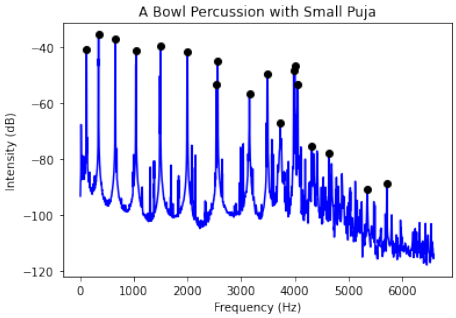

Figure B.1 A-Bowl frequency spectrum after striking with small puja

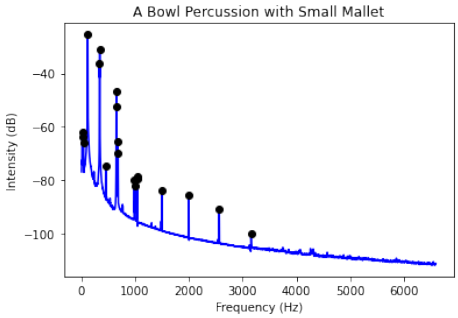

Figure B.2 A-Bowl frequency spectrum after striking with small mallet

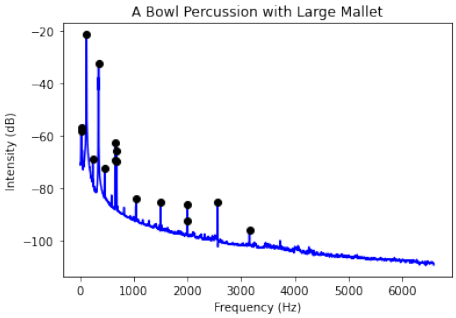

Figure B.3 A-Bowl frequency spectrum after striking with large mallet

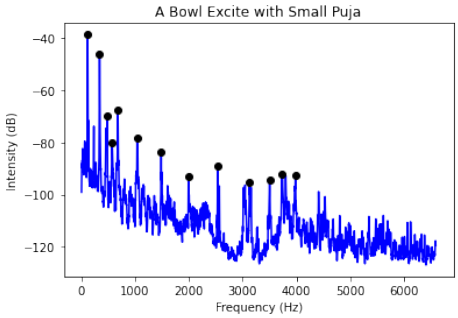

Figure B.4 A-Bowl frequency spectrum while continuously exciting with small puja

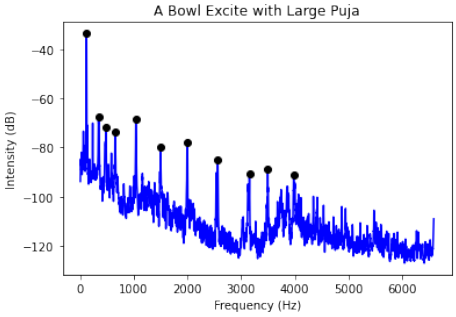

Figure B.5 A-Bowl frequency spectrum while continuously exciting with large puja

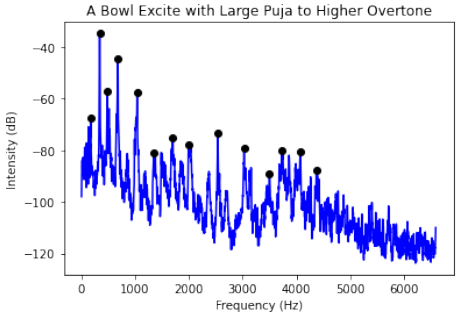

Figure B.6 A-Bowl frequency spectrum while continuously exciting to second overtone with large puja

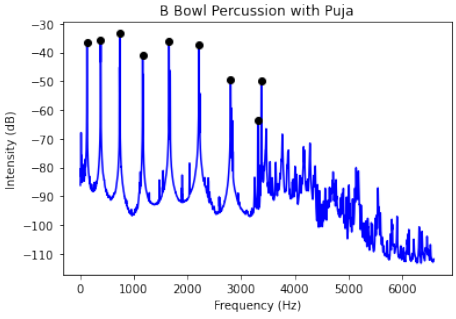

Figure B.7 B-Bowl frequency spectrum after striking with small puja

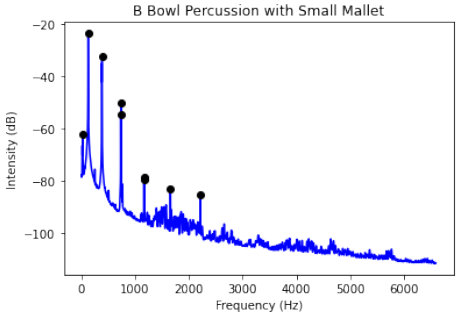

Figure B.8 B-Bowl frequency spectrum after striking with small mallet

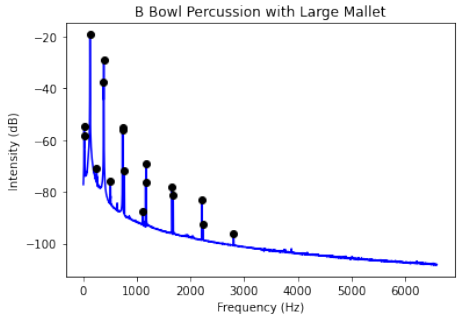

Figure B.9 B-Bowl frequency spectrum after striking with large mallet

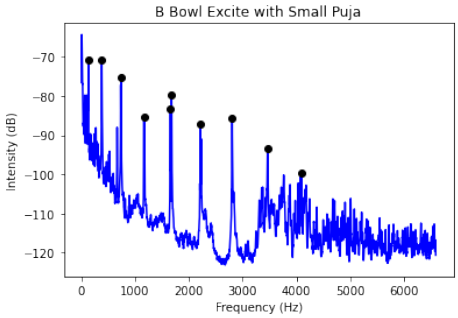

Figure B.10 B-Bowl frequency spectrum while continuously exciting with small puja

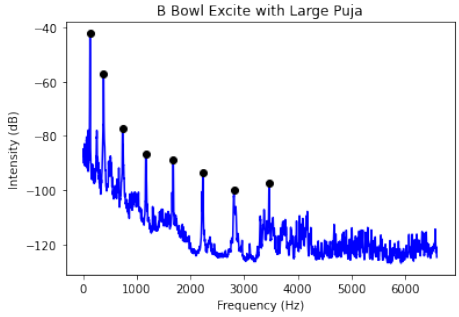

Figure B.11 B-Bowl frequency spectrum while continuously exciting with large puja

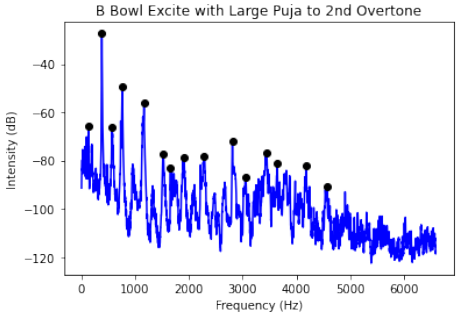

Figure B.12 B-Bowl frequency spectrum while continuously exciting to second overtone with large puja

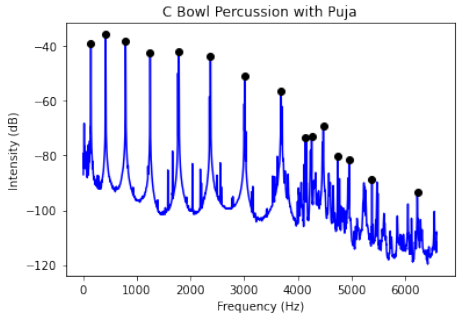

Figure B.13 C-Bowl frequency spectrum after striking with small puja

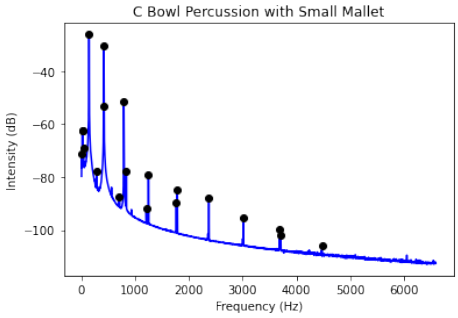

Figure B.14 C-Bowl frequency spectrum after striking with small mallet

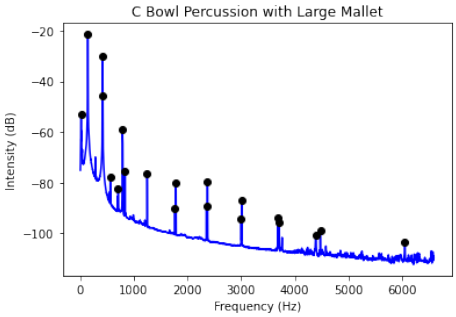

Figure B.15 C-Bowl frequency spectrum after striking with large mallet

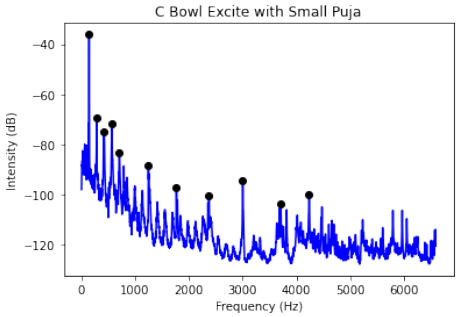

Figure B.16 C-Bowl frequency spectrum while continuously exciting with small puja

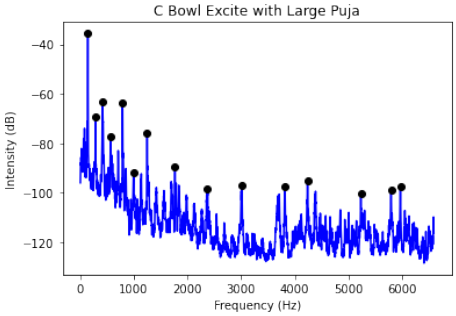

Figure B.17 C-Bowl frequency spectrum while continuously exciting with large puja

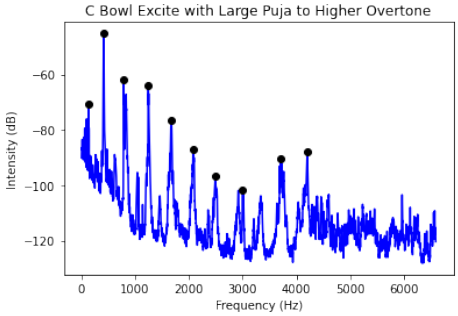

Figure B.18 C-Bowl frequency spectrum while continuously exciting to second overtone with large puja

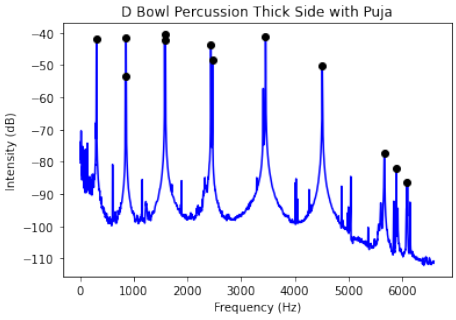

Figure B.19 D-Bowl frequency spectrum after striking the thick side with small puja

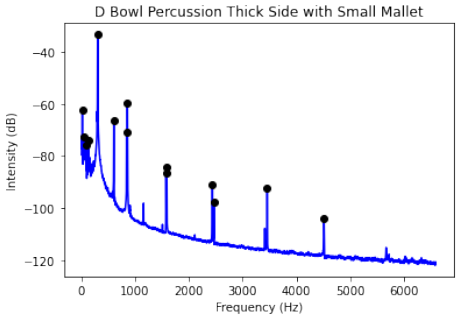

Figure B.20 D-Bowl frequency spectrum after striking the thick side with small mallet

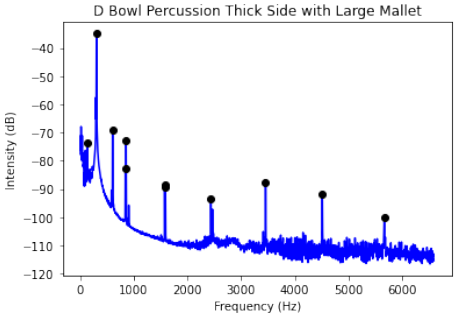

Figure B.21 D-Bowl frequency spectrum after striking the thick side with large mallet

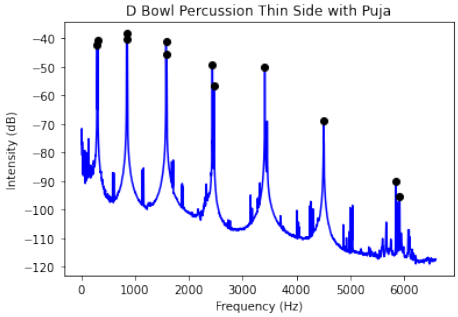

Figure B.22 D-Bowl frequency spectrum after striking the thin side with small puja

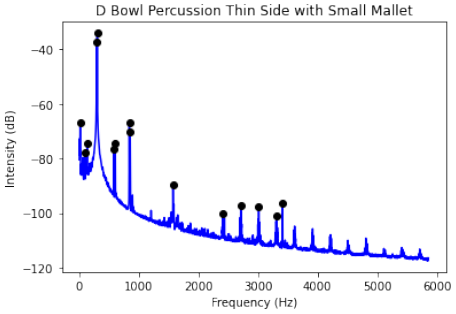

Figure B.23 D-Bowl frequency spectrum after striking the thin side with small mallet

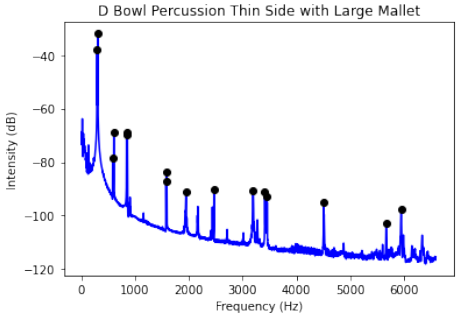

Figure B.24 D-Bowl frequency spectrum after striking the thin side with large mallet

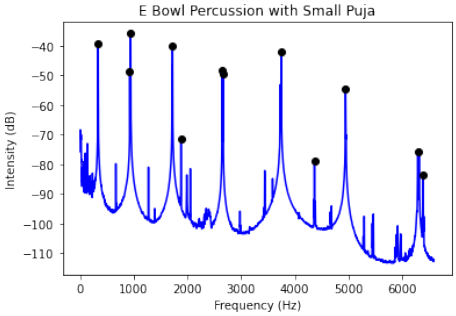

Figure B.25 E-Bowl frequency spectrum after striking with small puja

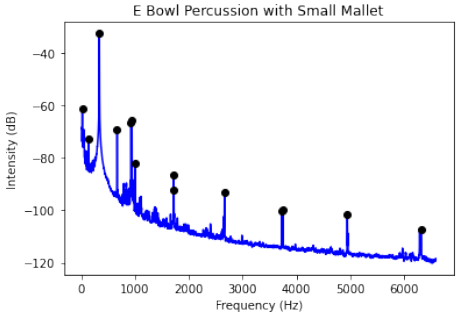

Figure B.26 E-Bowl frequency spectrum after striking with small mallet

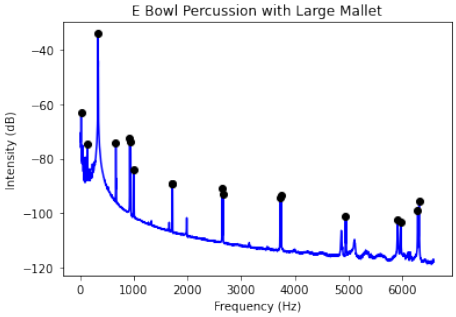

Figure B.27 E-Bowl frequency spectrum after striking with large mallet

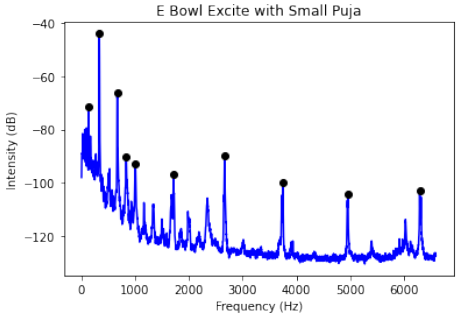

Figure B.28 E-Bowl frequency spectrum while continuously exciting with small puja

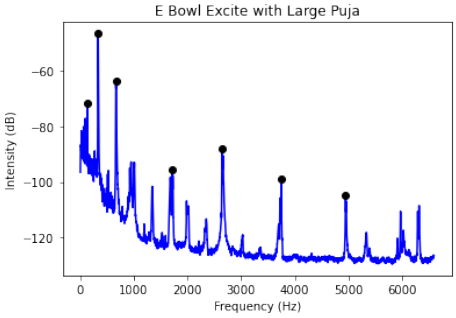

Figure B.29 E-Bowl frequency spectrum while continuously exciting with large puja

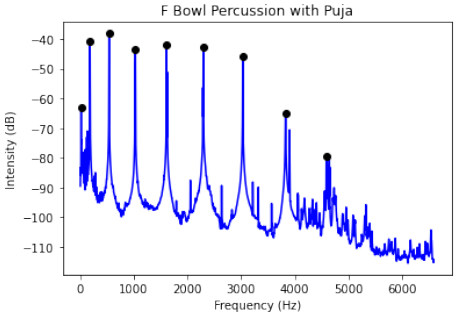

Figure B.30 F-Bowl frequency spectrum after striking with small puja

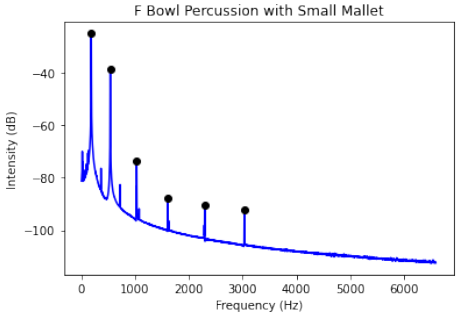

Figure B.31 F-Bowl frequency spectrum after striking with small mallet

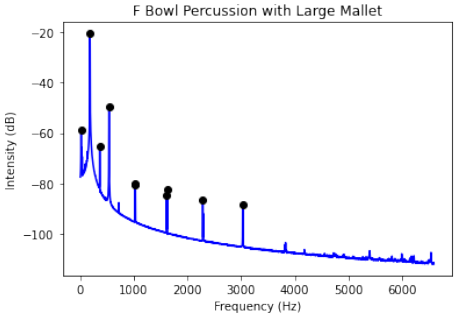

Figure B.32 F-Bowl frequency spectrum after striking with large mallet

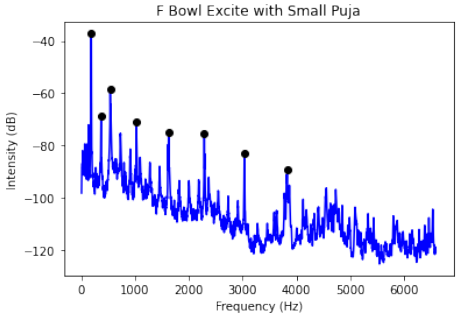

Figure B.33 F-Bowl frequency spectrum while continuously exciting with small puja

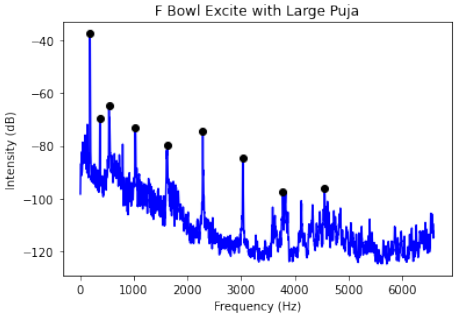

Figure B.34 F-Bowl frequency spectrum while continuously exciting with large puja

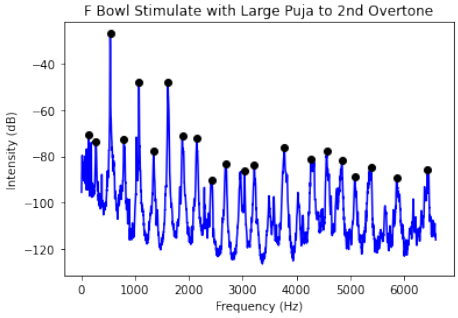

Figure B.35 F-Bowl frequency spectrum while continuously exciting to second overtone with large puja

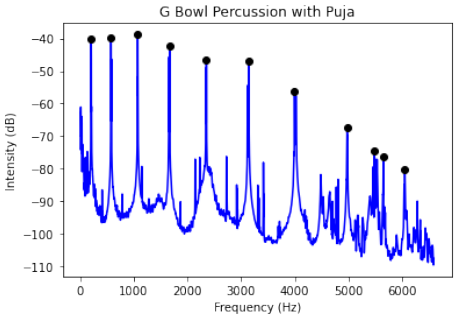

Figure B.36 G-Bowl frequency spectrum after striking with small puja

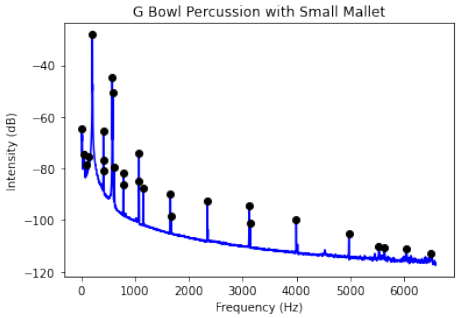

Figure B.37 G-Bowl frequency spectrum after striking with small mallet

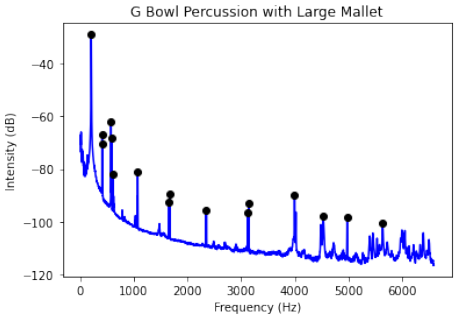

Figure B.38 G-Bowl frequency spectrum after striking with large mallet

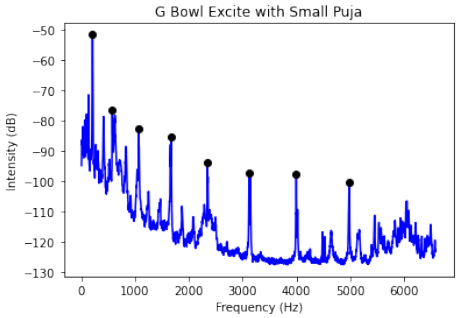

Figure B.39 G-Bowl frequency spectrum while continuously exciting with small puja

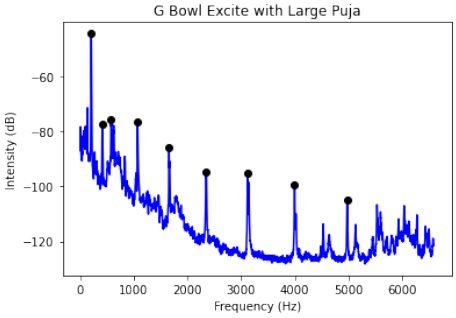

Figure B.40 G-Bowl frequency spectrum while continuously exciting with large puja

B.2 Vibrational Generator Frequency Spectrum Plots

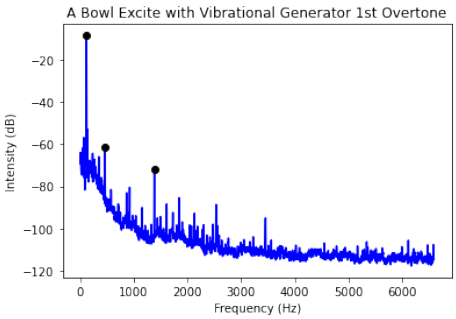

Figure B.41 A-Bowl frequency spectrum while exciting with vibrational generator to first overtone

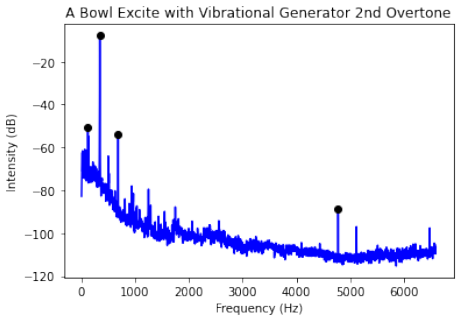

Figure B.42 A-Bowl frequency spectrum while exciting with vibrational generator to second overtone

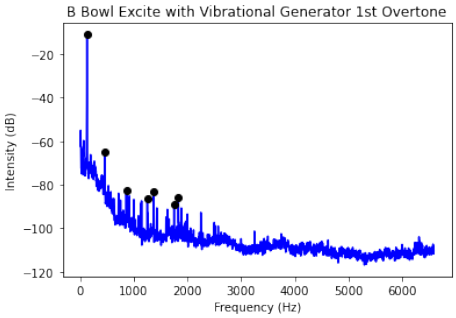

Figure B.43 B-Bowl frequency spectrum while exciting with vibrational generator to first overtone

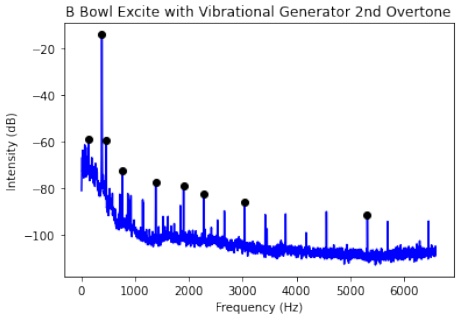

Figure B.44 B-Bowl frequency spectrum while exciting with vibrational generator to second overtone

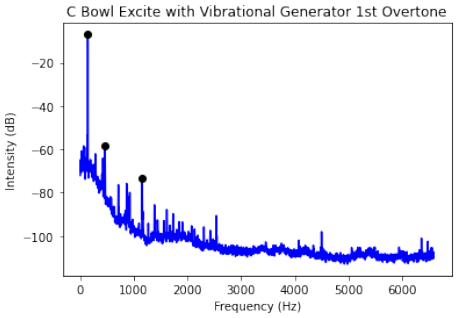

Figure B.45 C-Bowl frequency spectrum while exciting with vibrational generator to first overtone

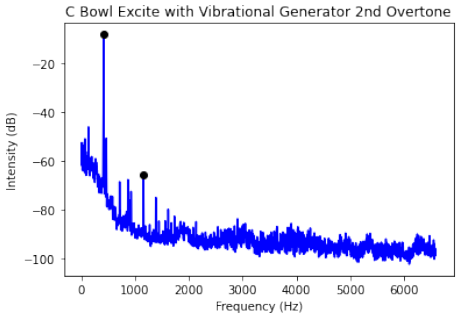

Figure B.46 C-Bowl frequency spectrum while exciting with vibrational generator to second overtone

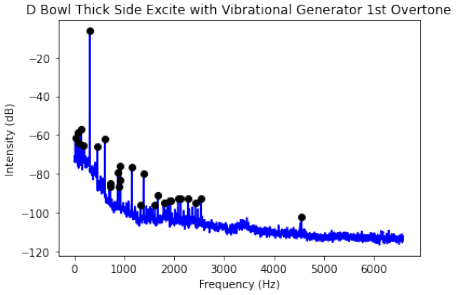

Figure B.47 D-Bowl frequency spectrum while exciting with vibrational generator to first overtone

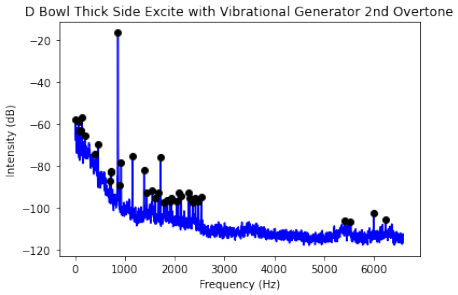

Figure B.48 D-Bowl frequency spectrum while exciting with vibrational generator to second overtone

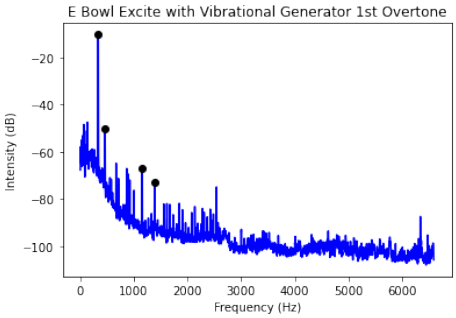

Figure B.49 E-Bowl frequency spectrum while exciting with vibrational generator to first overtone

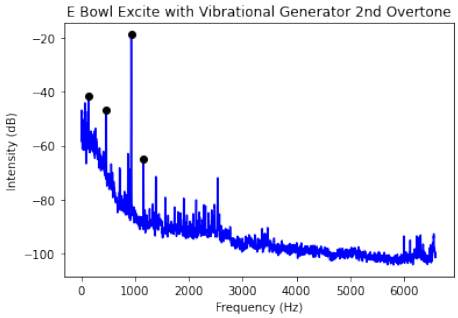

Figure B.50 E-Bowl frequency spectrum while exciting with vibrational generator to second overtone

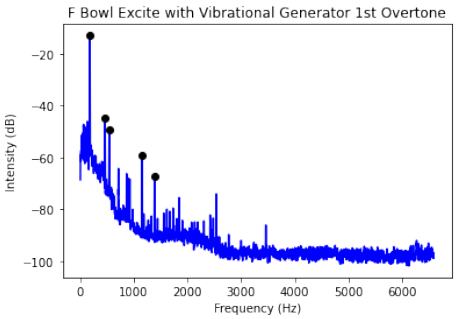

Figure B.51 F-Bowl frequency spectrum while exciting with vibrational generator to first overtone

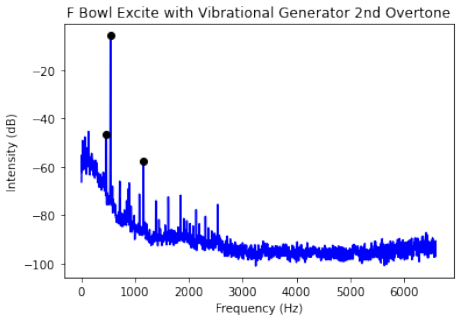

Figure B.52 F-Bowl frequency spectrum while exciting with vibrational generator to second overtone

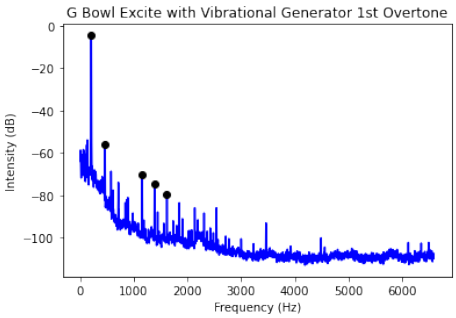

Figure B.53 G-Bowl frequency spectrum while exciting with vibrational generator to first overtone

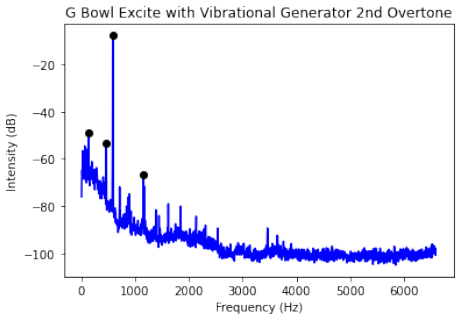

Figure B.54 G-Bowl frequency spectrum while exciting with vibrational generator to second overtone

B.3 Response Curves with Reduced Chi Squared Values

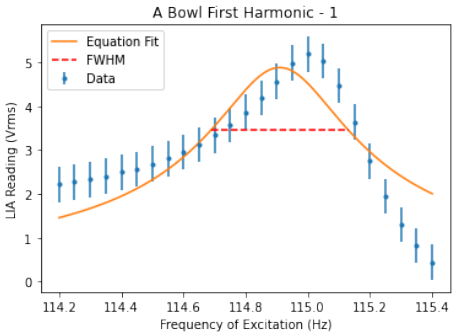

Figure B.55 A-Bowl response curve first harmonic first partial. RChiSq: 3.15

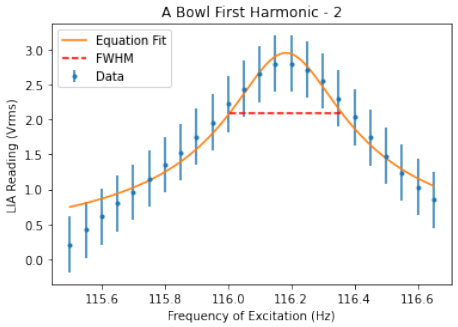

Figure B.56 A-Bowl response curve first harmonic second partial. RChiSq: 0.23

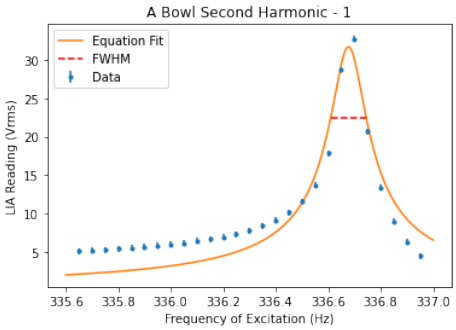

Figure B.57 A-Bowl response curve second harmonic first partial. RChiSq: 40.6

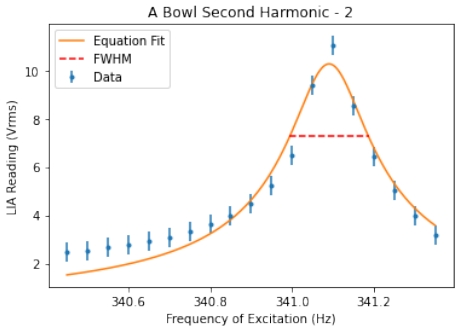

Figure B.58 A-Bowl response curve second harmonic second partial. RChiSq: 2.56

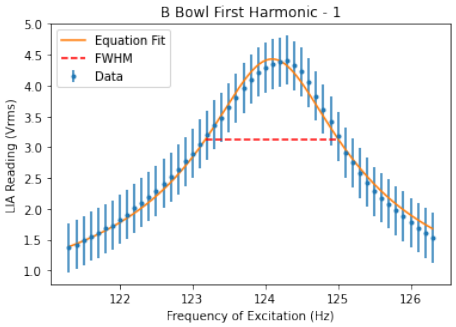

Figure B.59 B-Bowl response curve first harmonic first partial. RChiSq: 0.06

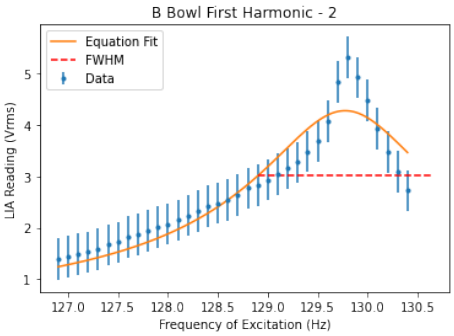

Figure B.60 B-Bowl response curve first harmonic second partial. RChiSq: 0.83

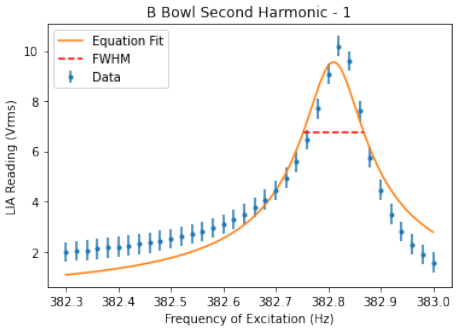

Figure B.61 B-Bowl response curve second harmonic first partial. RChiSq: 3.89

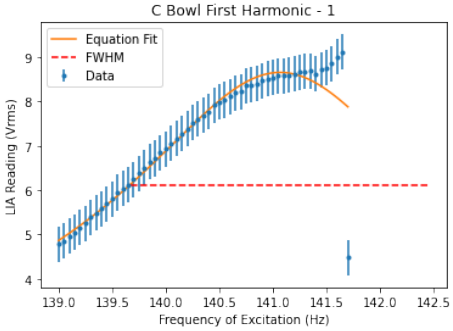

Figure B.62 C-Bowl response curve first harmonic first partial. RChiSq: 1.76

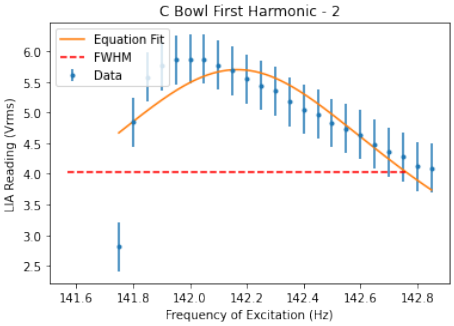

Figure B.63 C-Bowl response curve first harmonic second partial. RChiSq: 1.52

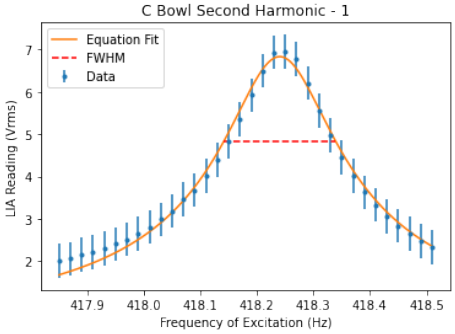

Figure B.64 C-Bowl response curve second harmonic first partial. RChiSq: 0.17

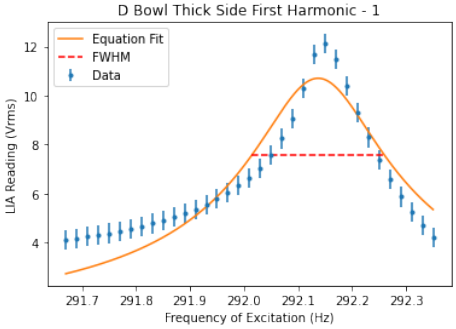

Figure B.65 D-Bowl thick side response curve first harmonic first partial. RChiSq: 5.28

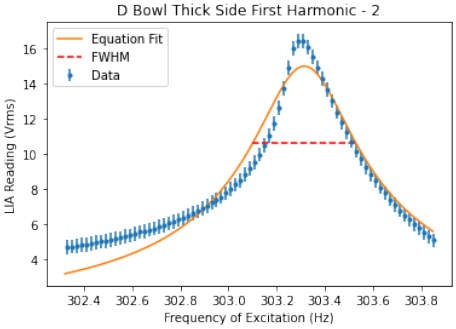

Figure B.66 D-Bowl thick side response curve first harmonic second partial. RChiSq: 4.87

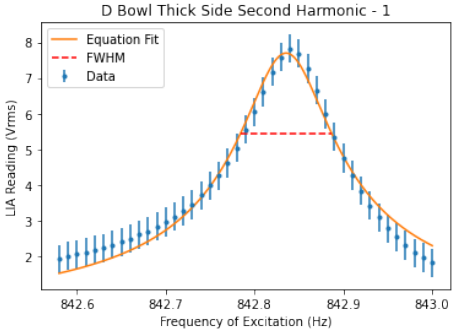

Figure B.67 D-Bowl thick side response curve second harmonic first partial. RChiSq: 0.56

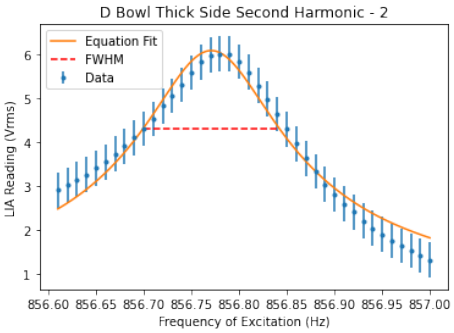

Figure B.68 D-Bowl thick side response curve second harmonic second partial. RChiSq: 0.50

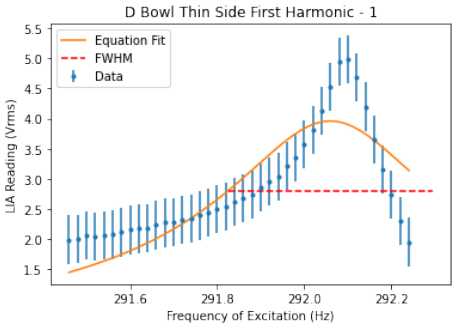

Figure B.69 D-Bowl thin side response curve first harmonic first partial. RChiSq: 1.61

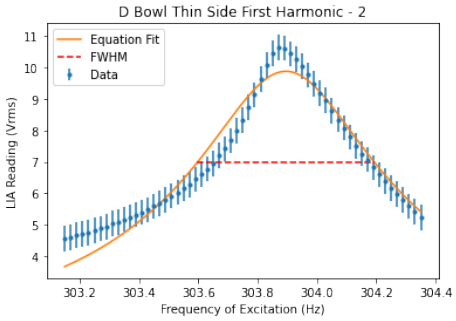

Figure B.70 D-Bowl thin side response curve first harmonic second partial. RChiSq: 1.33

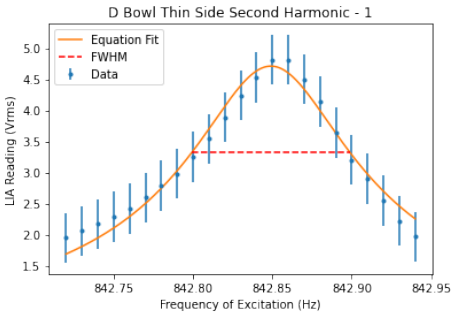

Figure B.71 D-Bowl thin side response curve second harmonic first partial. RChiSq: 0.18

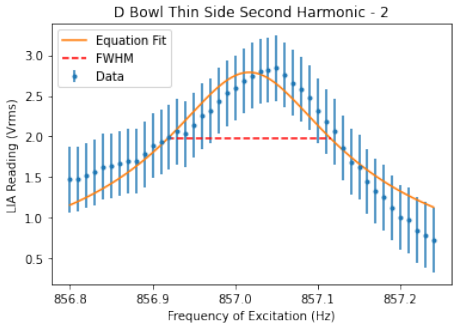

Figure B.72 D-Bowl thin side response curve second harmonic second partial. RChiSq: 0.27

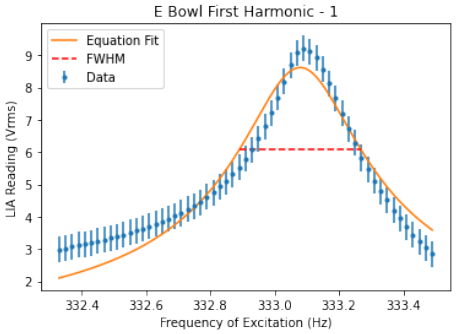

Figure B.73 E-Bowl response curve first harmonic first partial. RChiSq: 1.84

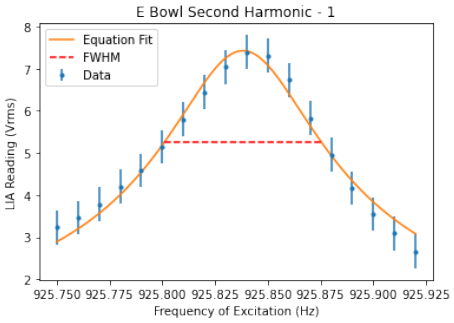

Figure B.74 E-Bowl response curve second harmonic first partial. RChiSq: 0.40

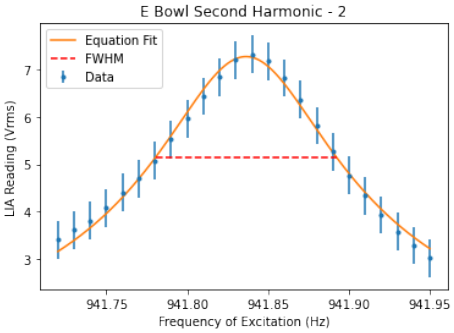

Figure B.75 E-Bowl response curve second harmonic second partial. RChiSq: 0.11

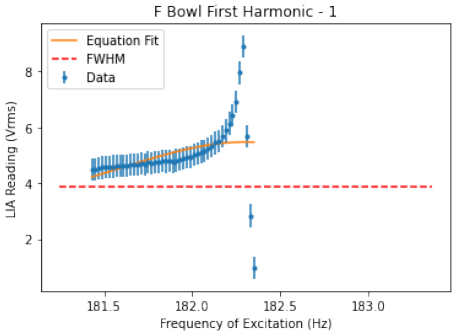

Figure B.76 F-Bowl response curve first harmonic first partial. RChiSq: 6.95

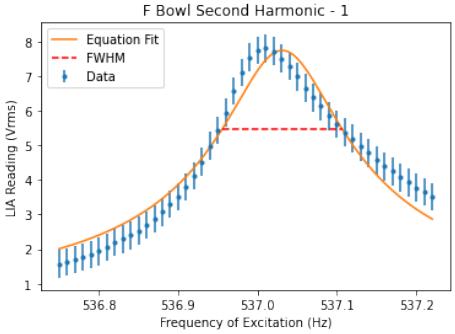

Figure B.77 F-Bowl response curve second harmonic first partial. RChiSq: 1.25

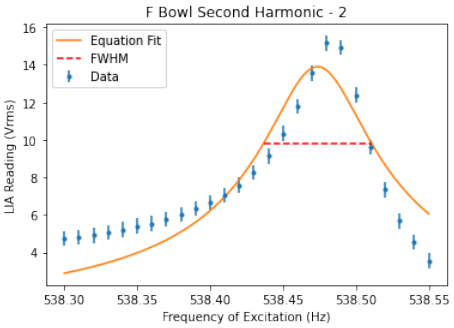

Figure B.78 F-Bowl response curve second harmonic second partial. RChiSq: 13.2

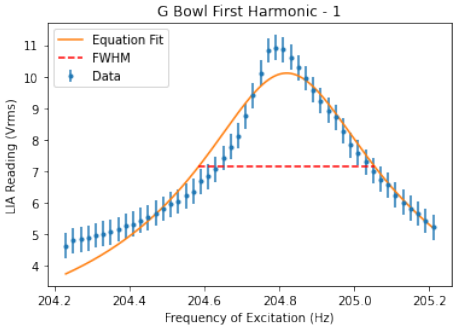

Figure B.79 G-Bowl response curve first harmonic first partial. RChiSq: 1.57

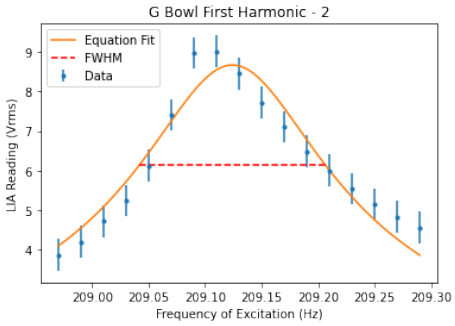

Figure B.80 G-Bowl response curve first harmonic second partial. RChiSq: 1.45

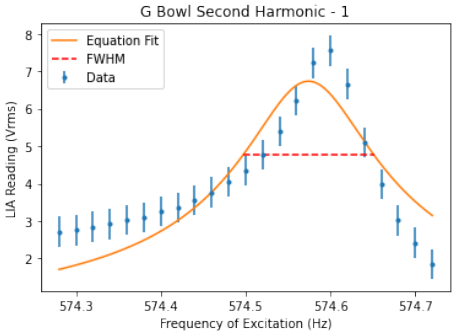

Figure B.81 G-Bowl response curve second harmonic first partial. RChiSq: 3.79

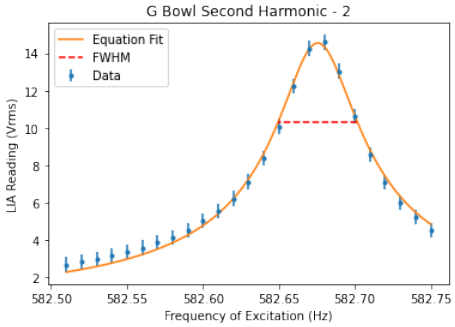

Figure B.82 G-Bowl response curve second harmonic second partial. RChiSq: 0.56

B.4 Nodal Mapping Plots

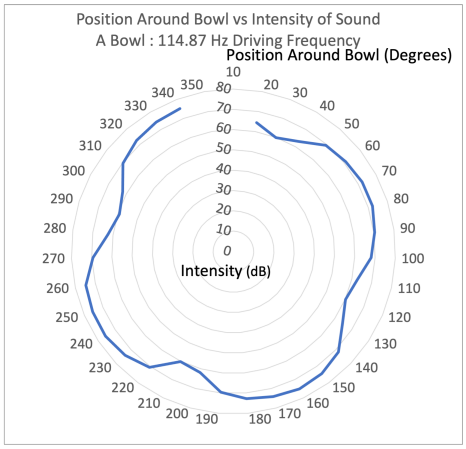

Figure B.83 A-Bowl first harmonic nodal map

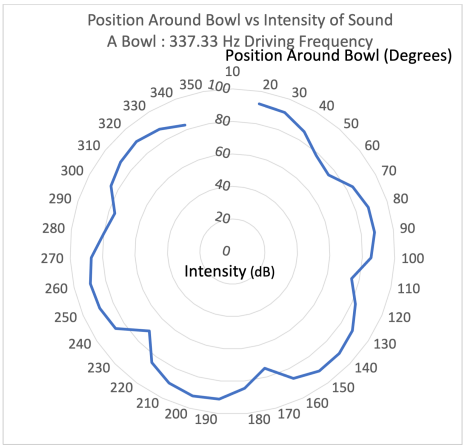

Figure B.84 A-Bowl second harmonic nodal map

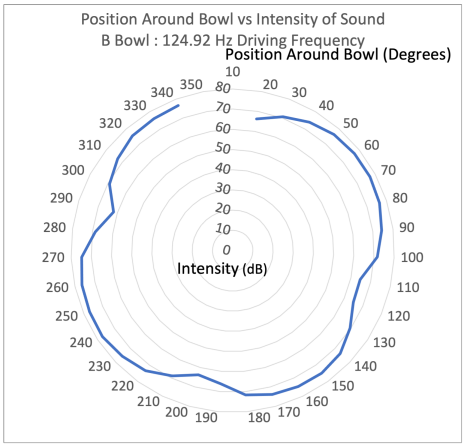

Figure B.85 B-Bowl first harmonic nodal map

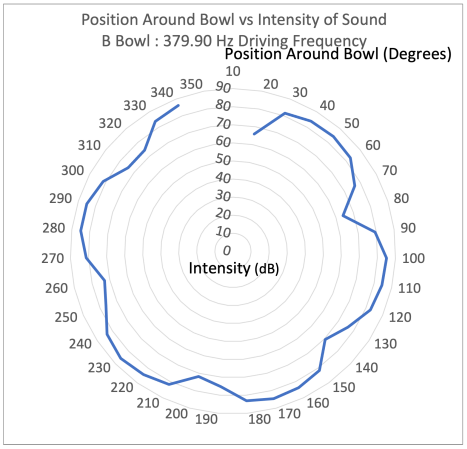

Figure B.86 B-Bowl second harmonic nodal map

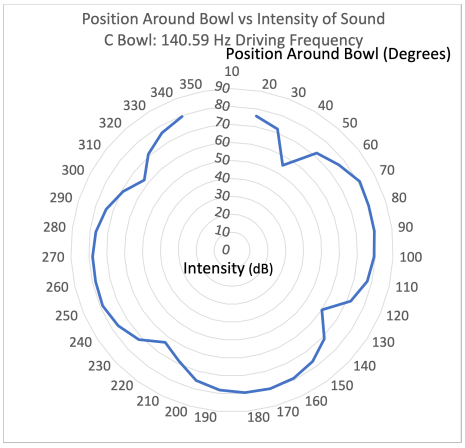

Figure B.87 C-Bowl first harmonic nodal map

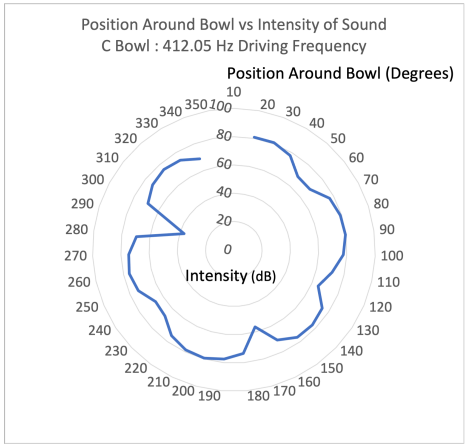

Figure B.88 C-Bowl second harmonic nodal map

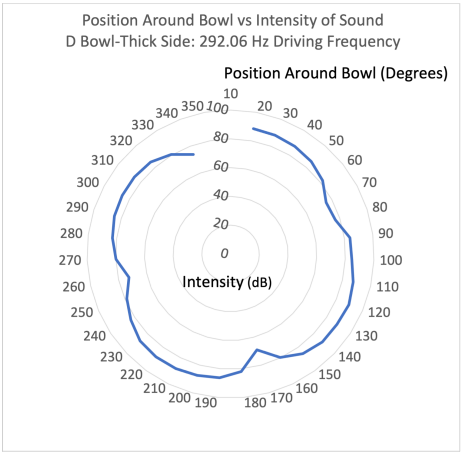

Figure B.89 D-Bowl thick side first harmonic first partial nodal map

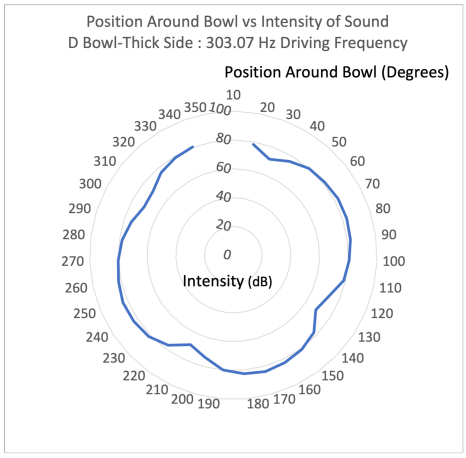

Figure B.90 D-Bowl thick side first harmonic second partial nodal map

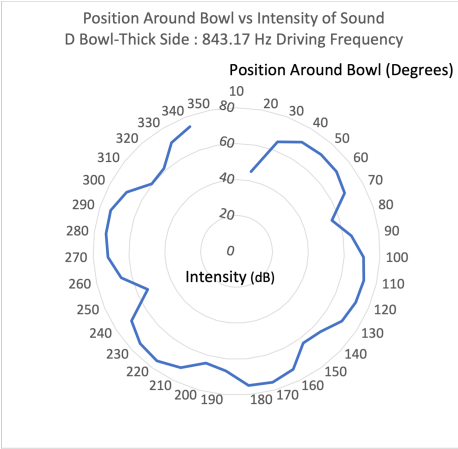

Figure B.91 D-Bowl thick side second harmonic nodal map

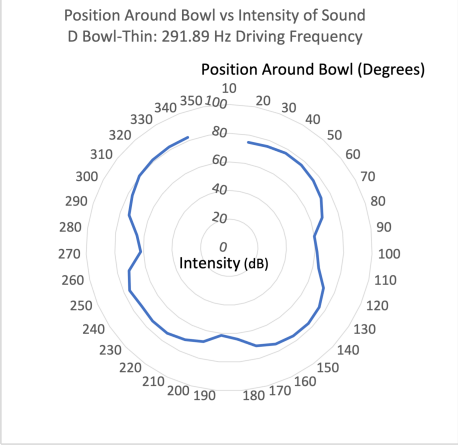

Figure B.92 D-Bowl thin side first harmonic first partial nodal map

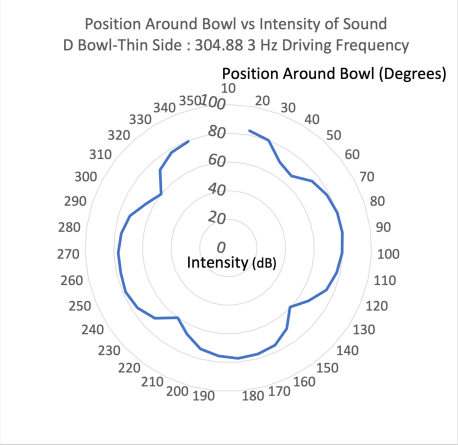

Figure B.93 D-Bowl thin side first harmonic second partial nodal map

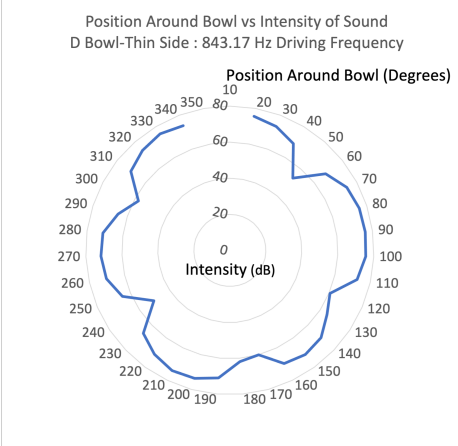

Figure B.94 D-Bowl thin side second harmonic nodal map

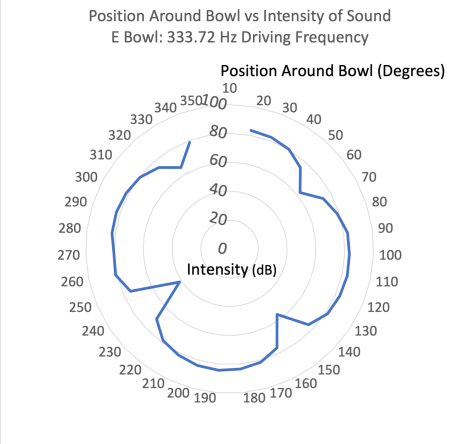

Figure B.95 E-Bowl first harmonic nodal map

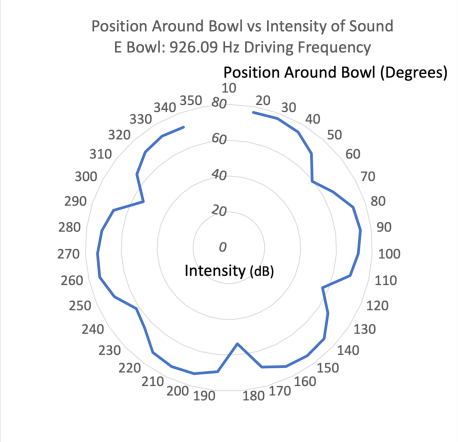

Figure B.96 E-Bowl second harmonic nodal map

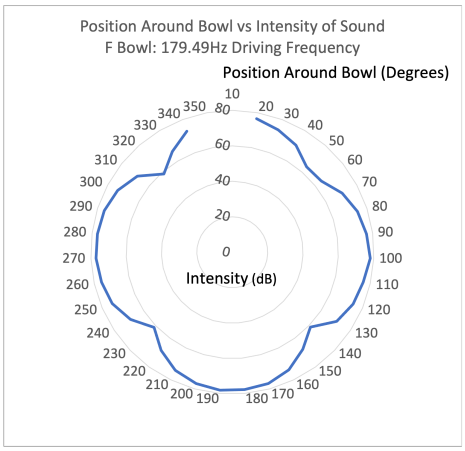

Figure B.97 F-Bowl First harmonic nodal map

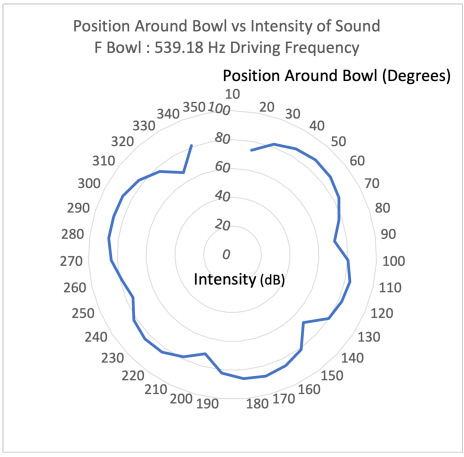

Figure B.98 F-Bowl second harmonic nodal map

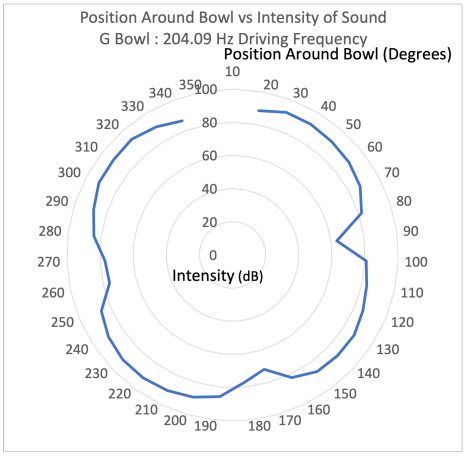

Figure B.99 G-Bowl first harmonic nodal map

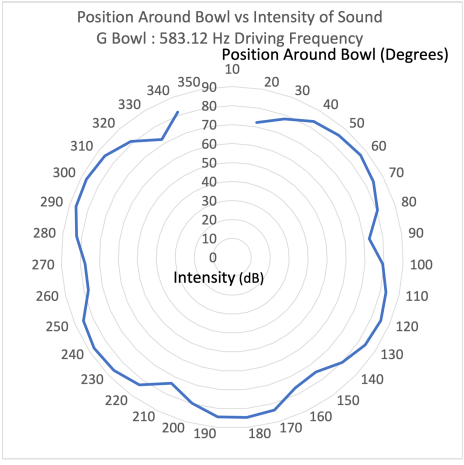

Figure B.100 G-Bowl second harmonic nodal map